TENAGA NASIONAL BERHAD

Reviewed / Verified
Engineer
Date : 14 SEP 23

GURSHARAN SINGH — DISTRIBUTION NETWORK DIVISION, TNB

Verification For Conformance With The Scheme And Functionality. This Verification Does Not Relieve The Contractor From Full Compliance With The Contractual Obligation

IMPORTANT

- Final approval of the detail schematic diagrams will be subjected to successful pre-commissioning test at site
- Approval is for secondary / primary equipments only
- There shall be no cost implication with this approval

TENAGA NASIONAL BERHAD

- [x] NO FURTHER COMMENTS
- [ ] REVISE AND RESUBMIT
- [ ] INFORMATION ONLY

By : GURSHARAN Date : 14/9/23

This endorsement shall not:
i. Be construed as acceptance of Variation and/or Contractor's Claim, and/or
ii. Relieve the Contractor from any Contractual obligations or responsibilities

PROFESSIONAL ENGINEER WITH PRACTICING CERTIFICATE
ELECTRONIC
Ir. AZHARUDDIN M KASSIM
C116116
MALAYSIA

ENGINEERING SERVICES DISEMAK 21 SEP 2023 DISTRIBUTION NETWORK DIVISION

Disemak & Didapati Teratur Mengikut
Spesifikasi Teknikal TNB Bagi Tujuan : Product Inspection

Jurutera Utama (Telecontrol & Telecommunication)
Engineering Services Unit
Distribution Network Division, TNB

Tarikh : 21 SEP 23

MOHD HAZIRIE — DISTRIBUTION NETWORK DIVISION, TNB

- [ ] For PRELIMINARY REVIEW
- [x] For CLIENT APPROVAL
- [ ] Issue For CONSTRUCTION
- [ ] Revised "AS-MANUFACTURED" (After 'FAT')
- [ ] Updated "AS-BUILT" (After 'SAT')

REMARKS:

Revision	Description	Date	Approved

DRAWN BY:	KHAIRI	DATE:	17.07.23
DESIGNED BY:	FIQRI	DATE:	17.07.23
CHECKED BY:	NAZMI	DATE:	17.07.23
REVIEWED BY:	AZMAN	DATE:	17.07.23

OWNER/CLIENT: TENAGA NASIONAL BERHAD

OWNER ENGINEER: TENAGA NASIONAL BERHAD

CONTRACTOR: Viscon

SUPPLIER/MANUFACTURER: Viscon

PROJECT TITLE: PEMBEKALAN, PEMASANGAN DAN PENGUJIAN DIGITAL OUTPUT CARD (DO), CONTROL PROTECTION BOARD (CPB) DAN DC SURGE PROTECTOR DEVICE BAGI RTU VISCON DUA C3 DI PENCAWANG DISTRIBUTION NETWORK

CONTRACT/PO NO: CONTRACT NO: TNB 1799/2023

SUBSTATION: PENCAWANG ELEKTRIK 11kV

DRAWING TITLE: RTU INTERNAL WIRING DRAWING

= PE	+ RTU	/

DRAWING NO.	REV.	
PDSB UPG1818 RIW	00	THIS SHEET: 3-1-1
		NEXT SHEET: 3-2-1

DATE	SHEET NO.	REVISION NO.	DESCRIPTION OF CHANGE

NO.	TITLE	DRG NO.	SHEET NO.	REV.	DATE	REMARKS
1	GENERAL	PDSB UPG1818 RIW				
1.1	- COVER SHEET	PDSB UPG1818 RIW	3-1-1	00	09.02.21	
1.2	- RIVISION HISTORY	PDSB UPG1818 RIW	3-2-1	00	09.02.21	
1.3	- TABLE OF CONTENTS 1	PDSB UPG1818 RIW	3-3-1	00	09.02.21	
1.4	- TABLE OF CONTENTS 2	PDSB UPG1818 RIW	3-3-2	00	09.02.21	
1.5	- TABLE OF LEGENDS	PDSB UPG1818 RIW	3-4-1	00	09.02.21	
1.6	- TABLE OF ABBREVIATIONS	PDSB UPG1818 RIW	3-4-2	00	09.02.21	
1.7	- DC DISTRIBUTION WIRING DIAGRAM	PDSB UPG1818 RIW	3-5-1	00	09.02.21	
1.8	- DC DISTRIBUTION WIRING DIAGRAM	PDSB UPG1818 RIW	3-5-2	00	09.02.21	
1.9	- AC DISTRIBUTION WIRING DIAGRAM	PDSB UPG1818 RIW	3-6-1	00	09.02.21	
1.10	- GROUNDING SYSTEM	PDSB UPG1818 RIW	3-7-1	00	09.02.21	
1.11	- GROUNDING SYSTEM	PDSB UPG1818 RIW	3-7-2	00	09.02.21	
1.12	- DIGITAL INPUT MODULE WIRING DIAGRAM - 1	PDSB UPG1818 RIW	3-8-1	00	09.02.21	
1.13	- DIGITAL INPUT MODULE WIRING DIAGRAM - 1	PDSB UPG1818 RIW	3-8-1A	00	09.02.21	
1.14	- DIGITAL INPUT MODULE WIRING DIAGRAM - 2	PDSB UPG1818 RIW	3-8-2	00	09.02.21	
1.15	- DIGITAL INPUT MODULE WIRING DIAGRAM - 2	PDSB UPG1818 RIW	3-8-2A	00	09.02.21	
1.16	- ANALOG INPUT MODULE WIRING DIAGRAM - 1	PDSB UPG1818 RIW	3-9-1	00	09.02.21	
1.17	- ANALOG INPUT MODULE WIRING DIAGRAM - 1	PDSB UPG1818 RIW	3-9-1A	00	09.02.21	
1.18	- ANALOG INPUT MODULE WIRING DIAGRAM - 2	PDSB UPG1818 RIW	3-9-2	00	09.02.21	
1.19	- ANALOG INPUT MODULE WIRING DIAGRAM - 2	PDSB UPG1818 RIW	3-9-2A	00	09.02.21	
1.20	- DIGITAL OUTPUT MODULE WIRING DIAGRAM - 1	PDSB UPG1818 RIW	3-10-1	00	09.02.21	
1.21	- 30V (RMU TYPE) - CONTROL FLOW DIAGRAM	PDSB UPG1818 RIW	3-10-1A	00	09.02.21	
1.22	- 30V (RMU TYPE) - BEFORE MODIFICATION WIRING DIAGRAM	PDSB UPG1818 RIW	3-10-1B	00	09.02.21	
1.23	- 30V (RMU TYPE) - AFTER MODIFICATION WIRING DIAGRAM	PDSB UPG1818 RIW	3-10-1C	00	09.02.21	
1.24	- 30V (VCB TYPE) - CONTROL FLOW DIAGRAM	PDSB UPG1818 RIW	3-10-1D	00	09.02.21	
1.25	- 30V (VCB TYPE) - BEFORE MODIFICATION WIRING DIAGRAM	PDSB UPG1818 RIW	3-10-1E	00	09.02.21	
1.26	- 30V (VCB TYPE) - AFTER MODIFICATION WIRING DIAGRAM	PDSB UPG1818 RIW	3-10-1F	00	09.02.21	

ENGINEERING SERVICES DISEMAK 21 SEP 2023 DISTRIBUTION NETWORK DIVISION

GURSHARAN SINGH 10 AUG 23 ES DISTRIBUTION NETWORK DIVISION, TNB

REV.	DATE	DRN/CHK/REV		
			DATE	17.07.23
			DRAWN	KHAIRI
			CHECKED	NAZMI
			REVISED	AZMAN

CLIENT: TENAGA NASIONAL BERHAD

CONTRACT NO: TNB 1799/2023

CONTRACTOR: Viscon

SUP./MFR.: Viscon

TITLE: TELECONTROL OVERVIEW DRAWING

TABLE OF CONTENTS

SUBSTATION: PENCAWANG ELEKTRIK 11kV

= PE | + RTU | /

DRAWING NO.: PDSB UPG1818 RIW

REV.: 00

THIS SHEET: 3-3-1

NEXT SHEET: 3-4-1

NO.	TITLE	DRG NO.	SHEET NO.	REV.	DATE	REMARKS
1.27	- DIGITAL OUTPUT MODULE WIRING DIAGRAM - 2	PDSB UPG1818 RIW	3-10-2	00	09.02.21	
1.28	- DIGITAL OUTPUT MODULE WIRING DIAGRAM - 2	PDSB UPG1818 RIW	3-10-2A	00	09.02.21	
1.29	- CONTROL ENABLE SWITCH CIRCUIT DIAGRAM	PDSB UPG1818 RIW	3-11-1	00	09.02.21	
1.30	- CONTROL ENABLE SWITCH CIRCUIT DIAGRAM	PDSB UPG1818 RIW	3-11-1A	00	09.02.21	
1.31	- CPB WIRING CIRCUIT	PDSB UPG1818 RIW	3-11-2	00	09.02.21	
1.32	- CPB WIRING CIRCUIT	PDSB UPG1818 RIW	3-11-2A	00	09.02.21	
1.33	- RTU COMMUNICATION MODULE WIRING - NCC	PDSB UPG1818 RIW	3-12-1	00	09.02.21	
1.34	- RTU COMMUNICATION WIRING - IED AND SATELLITE RTU	PDSB UPG1818 RIW	3-13-1	00	09.02.21	
1.35	- RTU INTERNAL SUBRACKS / INTER MODULE WIRING - CABINET	PDSB UPG1818 RIW	3-14-1	00	09.02.21	
1.36	- RTU DUMMY CB CIRCUIT	PDSB UPG1818 RIW	3-15-1	00	09.02.21	
1.37	- RTU FAULT ALARM CIRCUIT	PDSB UPG1818 RIW	3-16-1	00	09.02.21	
1.38	- RTU FAULT ALARM CIRCUIT	PDSB UPG1818 RIW	3-16-2	00	09.02.21	

ENGINEERING SERVICES DISEMAK 21 SEP 2023 DISTRIBUTION NETWORK DIVISION

GURSHARAN SINGH 10 AUG 23 ES DISTRIBUTION NETWORK DIVISION TNB

REV.	DATE	DRN/CHK/REV		
			DATE	17.07.23
			DRAWN	KHAIRI
			CHECKED	NAZMI
			REVISED	AZMAN

CLIENT: TENAGA NASIONAL BERHAD

CONTRACT NO: TNB 1799/2023

CONTRACTOR: Viscon

SUP./MFR.: Viscon

TITLE: TELECONTROL OVERVIEW DRAWING

TABLE OF CONTENTS

SUBSTATION: PENCAWANG ELEKTRIK 11kV

= PE | + RTU | /

DRAWING NO.: PDSB UPG1818 RIW

REV.: 00

THIS SHEET: 3-3-1

NEXT SHEET: 3-4-1

LEGEND	DESCRIPTIONS
	DOOR SWITCH (NORMALLY CLOSED)
	DRY CONTACT
	FILAMENT LIGHT BULB
	LEASED LINE MODEM
	MINIATURE CIRCUIT BREAKER (2-POLE)
	RS232 TO RS485 CONVERTER
	SURGE ARRESTOR
	WIRING TERMINAL BLOCK

LEGEND	DESCRIPTIONS



REV.	DATE	DRN/CHK/REV		
			DATE	17.07.23
			DRAWN	KHAIRI
			CHECKED	NAZMI
			REVISED	AZMAN

CLIENT: TENAGA NASIONAL BERHAD

CONTRACT NO: TNB 1799/2023

CONTRACTOR: Viscon

SUP./MFR.: Viscon

TITLE: TELECONTROL OVERVIEW DRAWING

TABLE OF CONTENTS

SUBSTATION: PENCAWANG ELEKTRIK 11kV

= PE | + RTU | /

DRAWING NO.: PDSB UPG1818 RIW

REV.: 00

THIS SHEET: 3-4-1

NEXT SHEET: 3-4-2

ABBREVIATIONS	DESCRIPTIONS
AC	ALTERNATING CURRENT
ACT	AC TERMINAL
ADAM 4522	RS232 / 485 CONVERTER
AI	ANALOG INPUT
ASDU	APPLICATION SERVICES DATA UNIT
BCD	BINARY CODED DECIMAL
CAASDU	COMMON ADDRESS OF ASDU
CDT	COMMUNICATION DISTRIBUTION TERMINAL
CES	CONTROL ENABLED SWITCH
CESDT	CES DISTRIBUTION TERMINAL
CPB	CONTROL PROTECTION BOARD
CTRL	CONTROL
DB	DOUBLE BIT
DBA	DUMMY BREAKER & ALARM
DC	DIRECT CURRENT
DCDT	DC DISTRIBUTION TERMINAL
DI	DIGITAL INPUT
DIL	DISTRIBUTED INTERPORT LINK
DO	DIGITAL OUTPUT
DRG	DRAWING
ESD	ELECTROSTATIC DISCHARGE
GIB	GPS INTERFACE BOX
GPS	GLOBAL POSITIONING SYSTEM
HX	HOST CONTROLLER
IOA	INFORMATION OBJECT ADDRESS
LA	LINK ADDRESS
LA-GPS	LIGHTNING ARRESTOR GPS

ABBREVIATIONS	DESCRIPTIONS
LTB	LAMP TERMINAL BLOCK
MC	MARSHALLING CUBICLE
MCB	MINIATURE CIRCUIT BREAKER
NCC	NETWORK CONTROL CENTER
PDIL	POWER DISTRIBUTION & INTERPORT LINK
PDU	PROGRAM DIAGNOSTIC UTULITY
PPU	PENCAWANG PEMBAHAGIAN UTAMA
PT	POINT
REV	REVISION
RTU	REMOTE TERMINAL UNIT
RX	RECEIVE
SA-DC	SURGE ARRESTOR FOR DC
SA-LL	SURGE ARRESTOR FOR LEASE LINE
SA-RS232	SURGE ARRESTOR FOR COMMUNICATION PORT
SB	SINGLE BIT
SEG	SEGMENT
SIP	SUPERVISORY INTERFACE PANEL
SOE	SEQUENCE OF EVENT
TB	TERMINAL BLOCK
TX	TRANSMIT

ENGINEERING SERVICES DISEMAK 21 SEP 2023 DISTRIBUTION NETWORK DIVISION

GURSHARAN SINGH 10 AUG 23 ES DISTRIBUTION NETWORK DIVISION, TNB

REV.	DATE	DRN/CHK/REV		
			DATE	17.07.23
			DRAWN	KHAIRI
			CHECKED	NAZMI
			REVISED	AZMAN

CLIENT: TENAGA NASIONAL BERHAD

CONTRACT NO: TNB 1799/2023

CONTRACTOR: Viscon

SUP./MFR.: Viscon

TITLE: TELECONTROL OVERVIEW DRAWING

TABLE OF ABBREVIATIONS

SUBSTATION: PENCAWANG ELEKTRIK 11kV

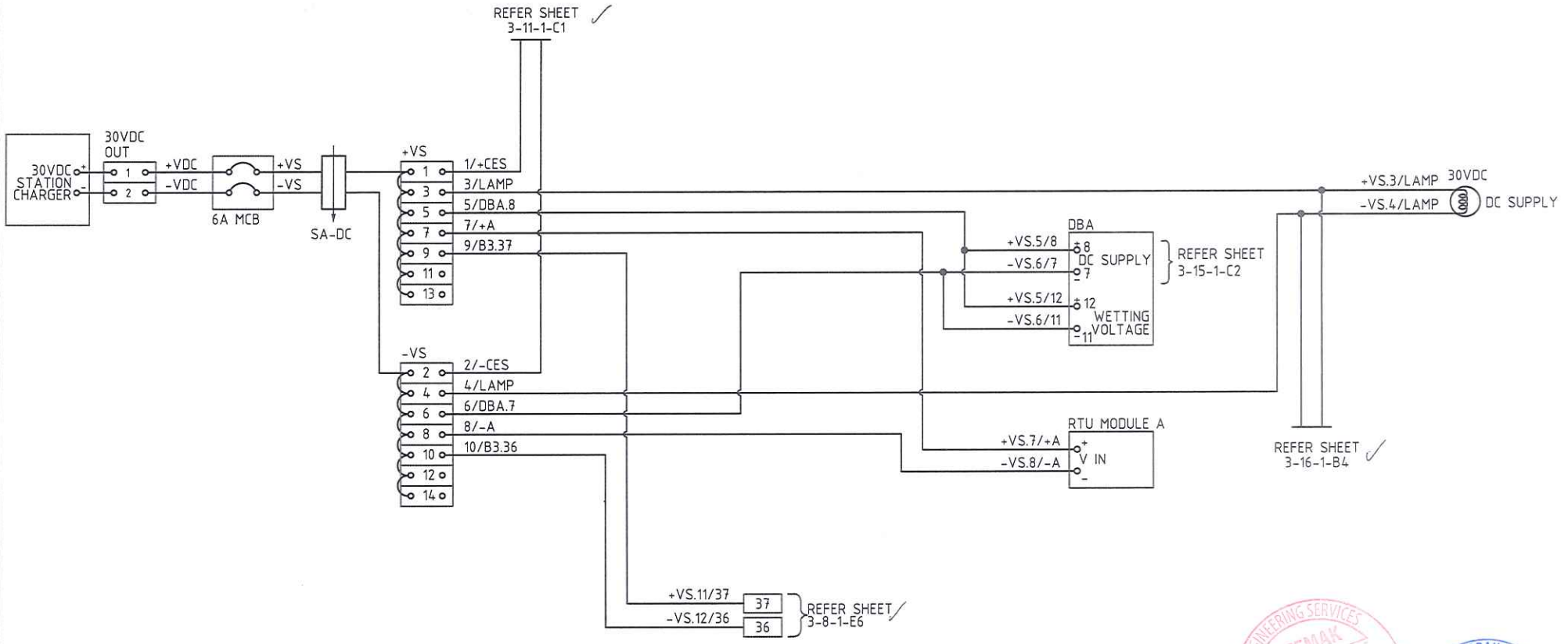
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DRAWING NO.: PDSB UPG1818 RIW

REV.: 00

THIS SHEET: 3-4-2

NEXT SHEET: 3-5-1

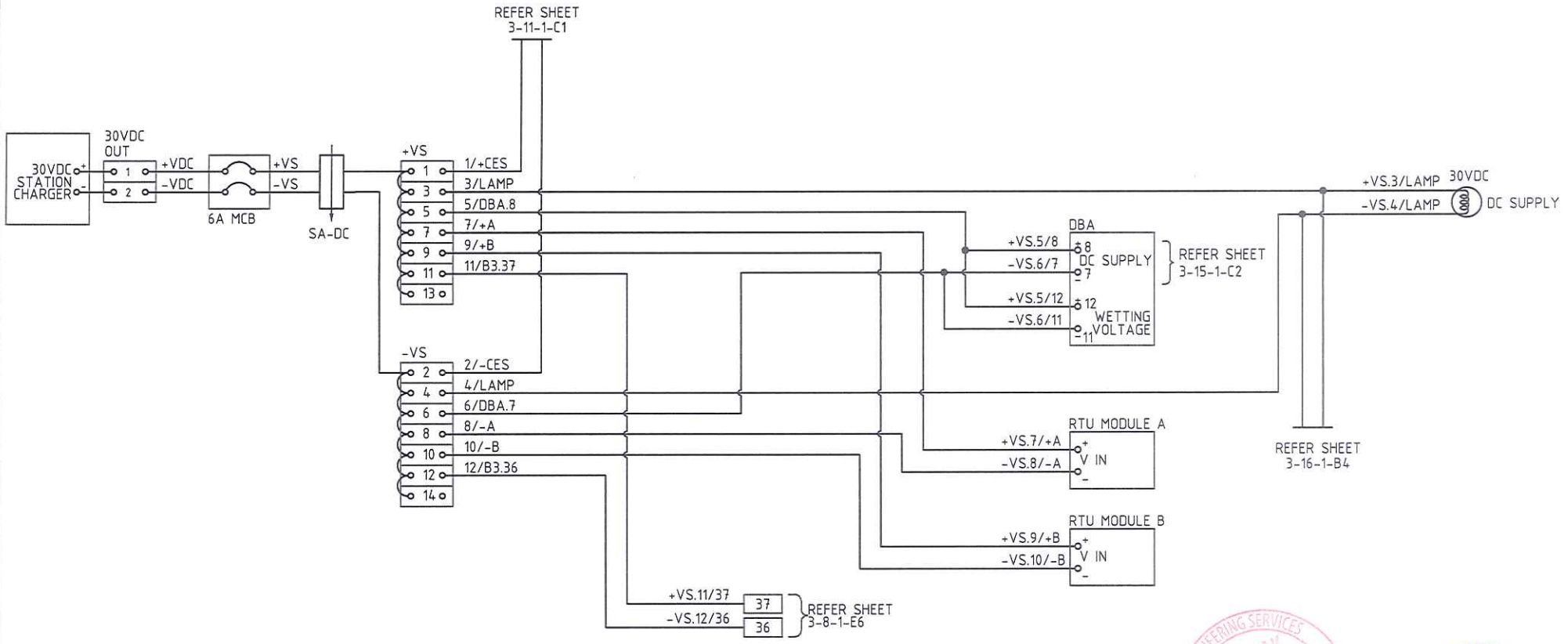


EXISTING DC DISTRIBUTION WIRING DIAGRAM (RMU TYPE)

ENGINEERING SERVICES DISEMAK 21 SEP 2023 DISTRIBUTION NETWORK DIVISION

GURSHARAN SINGH 10 AUG 23 ES DISTRIBUTION NETWORK DIVISION, TNB

			DATE	17.07.23	CLIENT: TENAGA NASIONAL BERHAD	CONTRACTOR: Viscon	TITLE: TELECONTROL OVERVIEW DRAWING	= PE	+ RTU /
			DRAWN	KHAIRI			DC DISTRIBUTION WIRING DIAGRAM		
			CHECKED	NAZMI	CONTRACT NO: TNB 1799/2023	SUP./MFR.: Viscon		DRAWING NO.: PDSB UPG1818 RIW	REV.: 00
REV.	DATE	DRN/CHK/REV	REVISED	AZMAN			SUBSTATION: PENCAWANG ELEKTRIK 11kV		THIS SHEET: 3-5-1 NEXT SHEET: 3-5-2

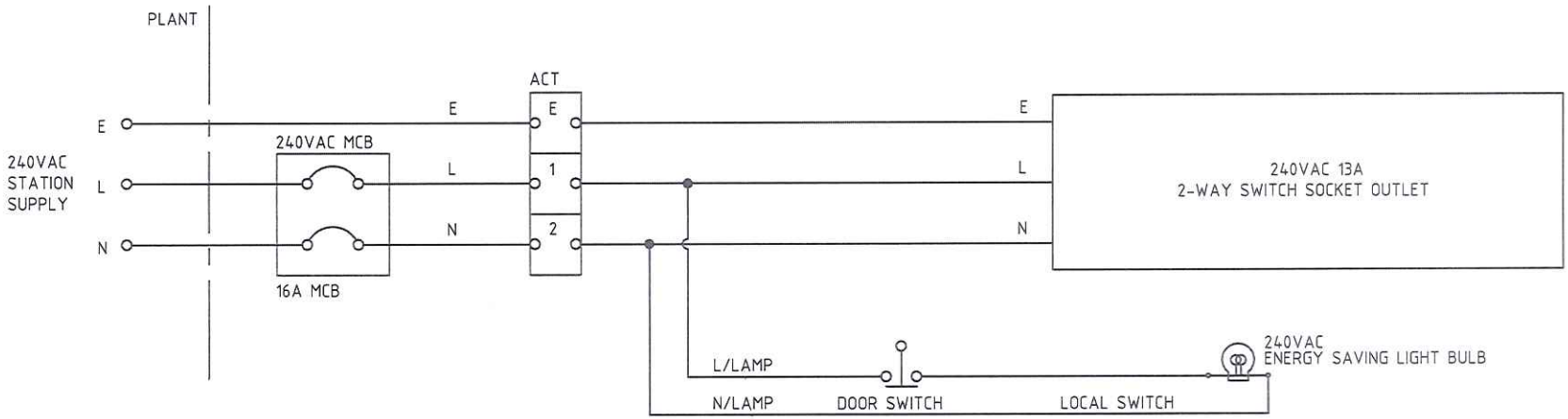


EXISTING DC DISTRIBUTION WIRING DIAGRAM (VCB TYPE)

ENGINEERING SERVICES DISEMAK 21 SEP 2023 DISTRIBUTION NETWORK DIVISION

GURSHARAN SINGH 10 AUG 23 ES DISTRIBUTION NETWORK DIVISION, TNB

			DATE	17.07.23	CLIENT: TENAGA NASIONAL BERHAD	CONTRACTOR: Viscon	TITLE: TELECONTROL OVERVIEW DRAWING	= PE	+ RTU	/	
			DRAWN	KHAIRI			DC DISTRIBUTION WIRING DIAGRAM				
			CHECKED	NAZMI	CONTRACT NO: TNB 1799/2023	SUP./MFR.: Viscon		DRAWING NO.: PDSB UPG1818 RIW	REV.: 00	THIS SHEET: 3-5-2	
REV.	DATE	DRN/CHK/REV	REVISED	AZMAN			SUBSTATION: PENCAWANG ELEKTRIK 11kV			NEXT SHEET: 3-6-1	



EXISTING AC DISTRIBUTION WIRING DIAGRAM

ENGINEERING SERVICES DISEMAK 21 SEP 2023 DISTRIBUTION NETWORK DIVISION

GURSHARAN SINGH * 10 AUG 23 ES DISTRIBUTION NETWORK DIVISION, TNB

REV.	DATE	DRN/CHK/REV		
			DATE	17.07.23
			DRAWN	KHAIRI
			CHECKED	NAZMI
			REVISED	AZMAN

CLIENT: TENAGA NASIONAL BERHAD

CONTRACT NO: TNB 1799/2023

CONTRACTOR: Viscon

SUP./MFR.: Viscon

TITLE: TELECONTROL OVERVIEW DRAWING
AC DISTRIBUTION WIRING DIAGRAM

SUBSTATION: PENCAWANG ELEKTRIK 11kV

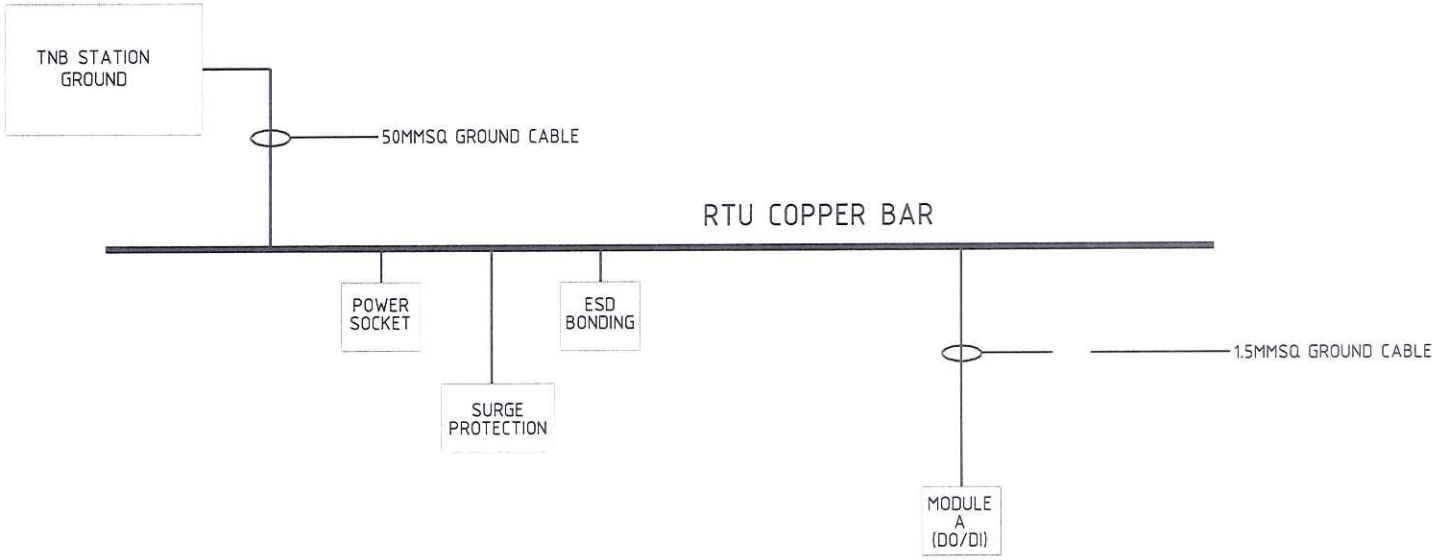
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REV.: 00

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NEXT SHEET: 3-7-1

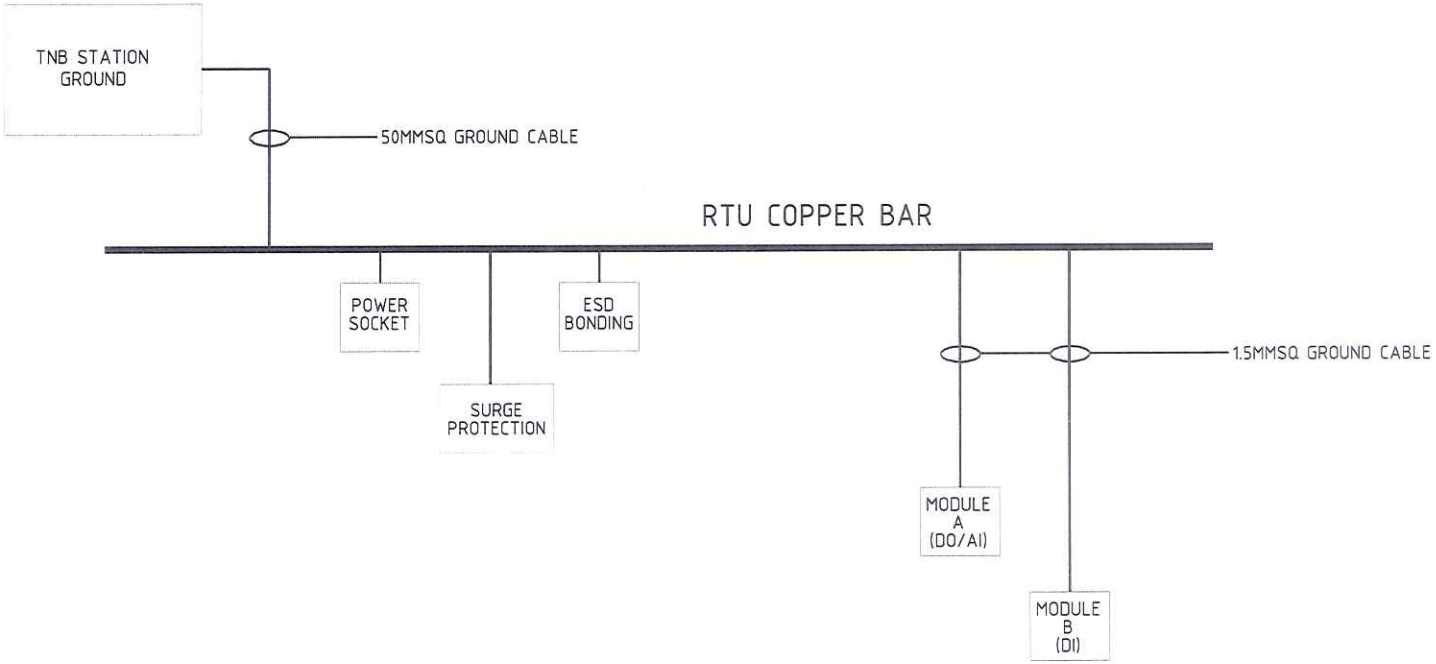


EXISTING GROUNDING SYSTEM (RMU TYPE)

ENGINEERING SERVICES DISEMAK 21 SEP 2023 DISTRIBUTION NETWORK DIVISION

GURSHARAN SINGH 10 AUG 23 ES DISTRIBUTION NETWORK DIVISION, TNB

			DATE	17.07.23	CLIENT: TENAGA NASIONAL BERHAD	CONTRACTOR: Viscon	TITLE: TELECONTROL OVERVIEW DRAWING	= PE	+ RTU /
			DRAWN	KHAIRI			GROUNDING SYSTEM		
			CHECKED	NAZMI	CONTRACT NO: TNB 1799/2023	SUP./MFR.: Viscon		DRAWING NO.: PDSB UPG1818 RIW	REV.: 00
REV.	DATE	DRN/CHK/REV	REVISED	AZMAN			SUBSTATION: PENCAWANG ELEKTRIK 11kV		THIS SHEET: 3-7-1 NEXT SHEET: 3-7-2

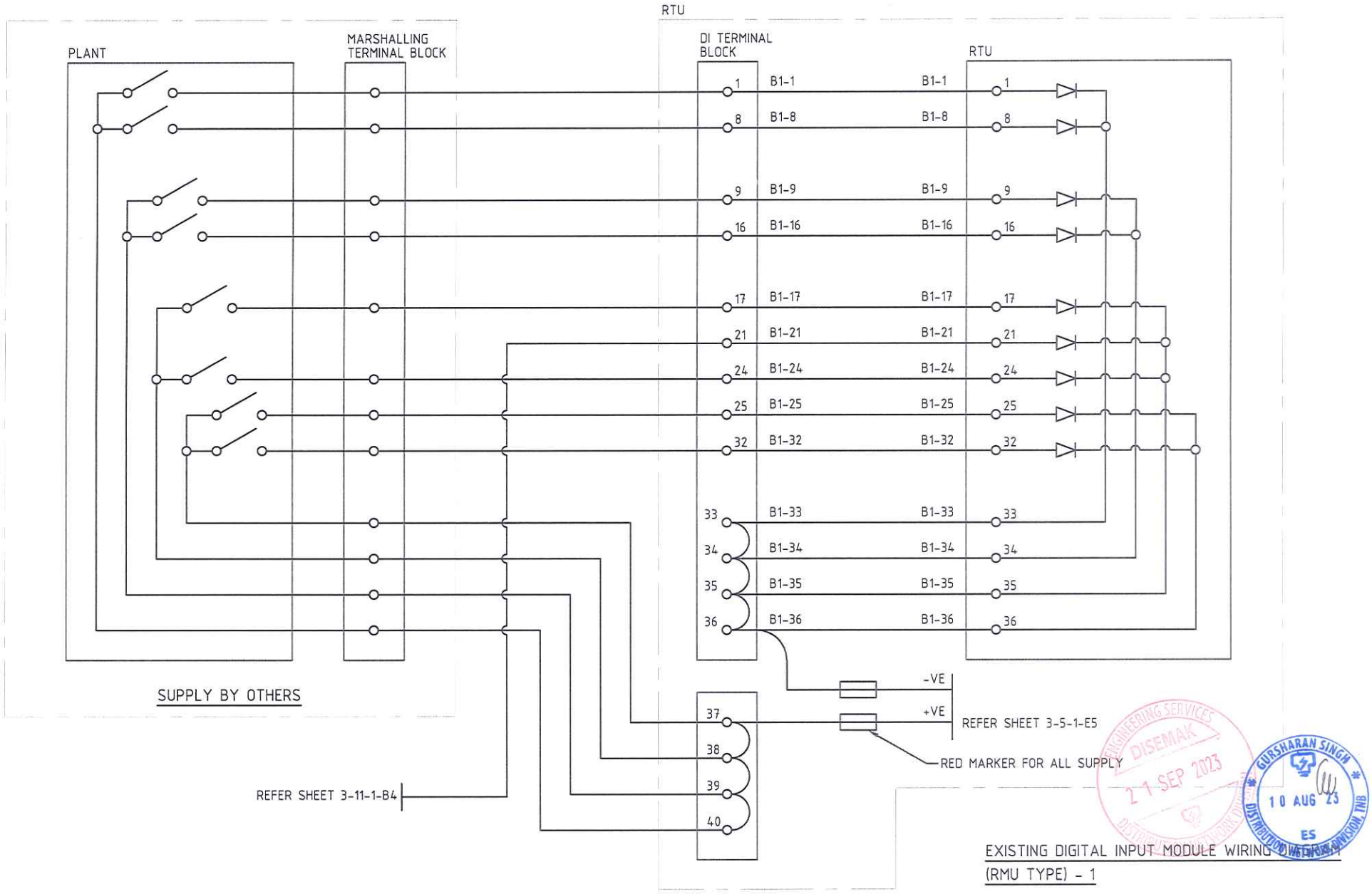


EXISTING GROUNDING SYSTEM (VCB TYPE)

ENGINEERING SERVICES DISEMAK 21 SEP 2023 DISTRIBUTION NETWORK DIVISION

GURSHARAN SINGH 10 AUG 23 ES DISTRIBUTION NETWORK DIVISION TNB

			DATE	17.07.23	CLIENT: TENAGA NASIONAL BERHAD	CONTRACTOR: Viscon	TITLE: TELECONTROL OVERVIEW DRAWING	= PE	+ RTU	/
			DRAWN	KHAIRI			GROUNDING SYSTEM			
			CHECKED	NAZMI	CONTRACT NO: TNB 1799/2023	SUP./MFR.: Viscon		DRAWING NO.: PDSB UPG1818 RIW	REV.: 00	THIS SHEET: 3-7-2
REV.	DATE	DRN/CHK/REV	REVISED	AZMAN			SUBSTATION: PENCAWANG ELEKTRIK 11kV			NEXT SHEET: 3-8-1

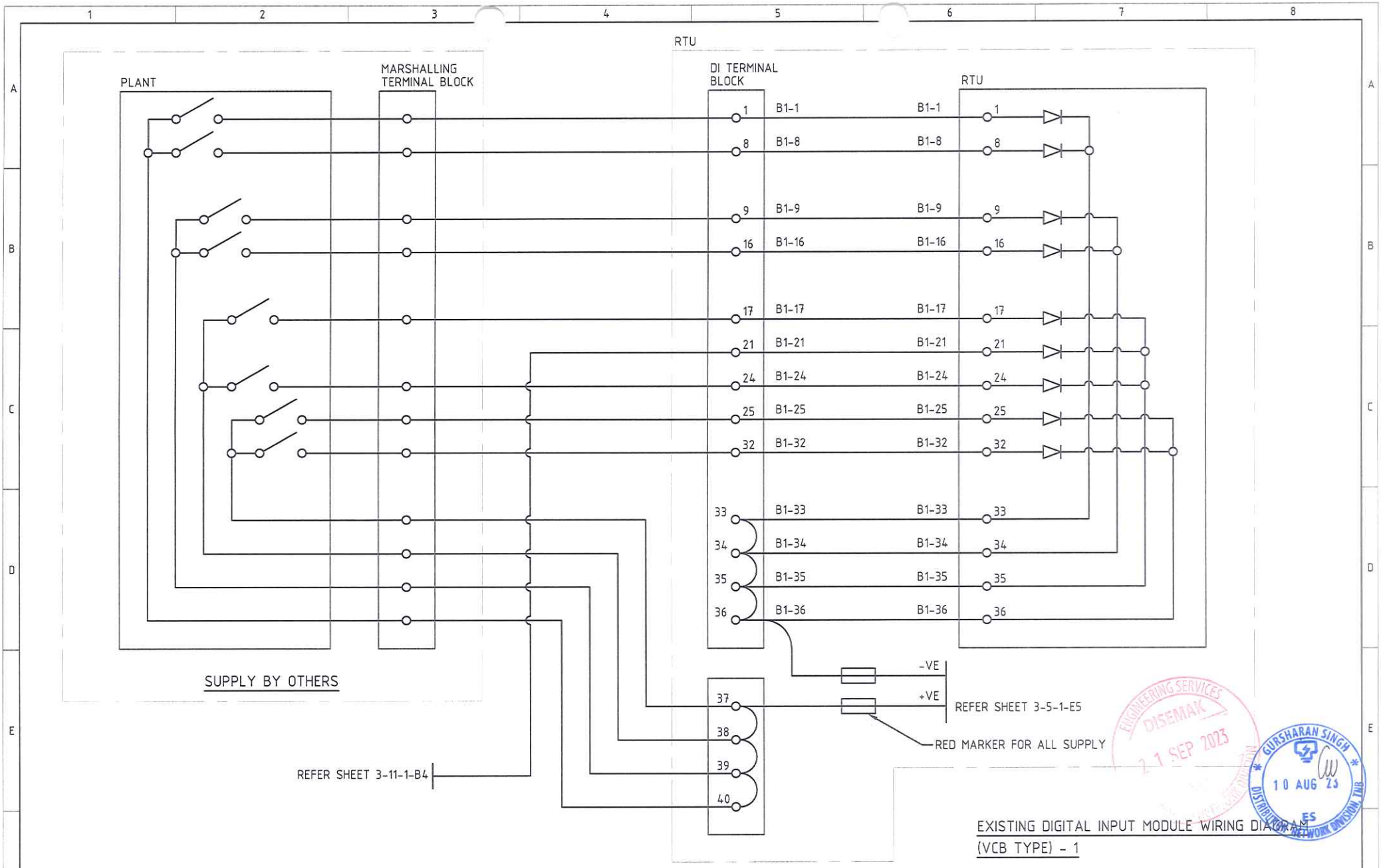


EXISTING DIGITAL INPUT MODULE WIRING DIAGRAM
(RMU TYPE) - 1

ENGINEERING SERVICES DISEMAK 21 SEP 2023 DISTRIBUTION NETWORK DIVISION

GURSHARAN SINGH 10 AUG 23 ES DISTRIBUTION DIVISION, TNB

			DATE	17.07.23	CLIENT: TENAGA NASIONAL BERHAD	CONTRACTOR: Viscon	TITLE: TELECONTROL OVERVIEW DRAWING	= PE	+ RTU /
			DRAWN	KHAIRI			DIGITAL INPUT MODULE WIRING DIAGRAM - 1		
			CHECKED	NAZMI	CONTRACT NO: TNB 1799/2023	SUP./MFR.: Viscon		DRAWING NO.: PDSB UPG1818 RIW	REV.: 00
REV.	DATE	DRN/CHK/REV	REVISED	AZMAN			SUBSTATION: PENCAWANG ELEKTRIK 11kV		THIS SHEET: 3-8-1 NEXT SHEET: 3-8-1A

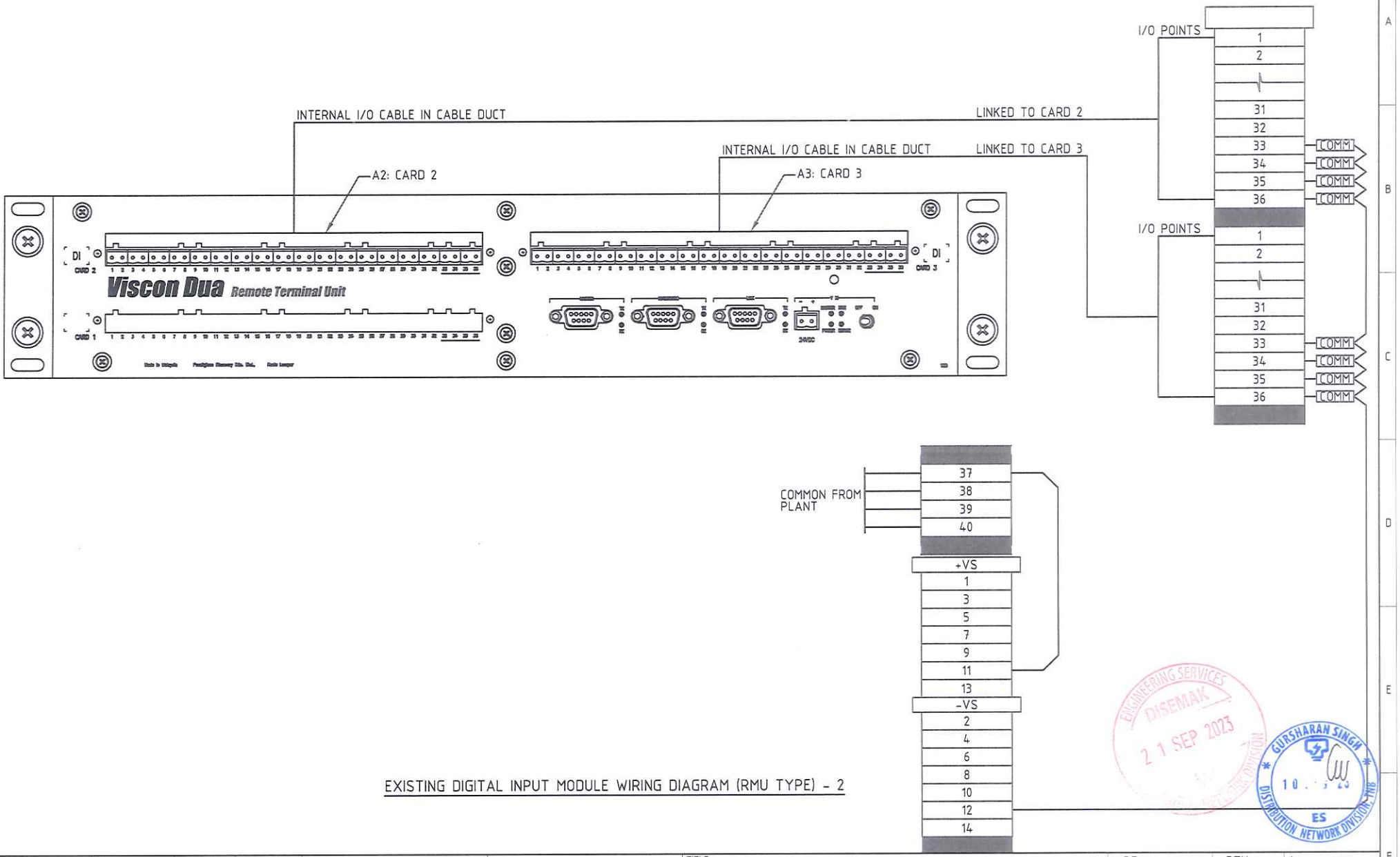


EXISTING DIGITAL INPUT MODULE WIRING DIAGRAM (VCB TYPE) - 1

ENGINEERING SERVICES DISEMAK 21 SEP 2023

GURSHARAN SINGH 10 AUG 23 ES DISTRIBUTION NETWORK DIVISION, TNB

			DATE	17.07.23	CLIENT: TENAGA NASIONAL BERHAD	CONTRACTOR: Viscon	TITLE: TELECONTROL OVERVIEW DRAWING	= PE	+ RTU /
			DRAWN	KHAIRI			DIGITAL INPUT MODULE WIRING DIAGRAM - 1		
			CHECKED	NAZMI	CONTRACT NO: TNB 1799/2023	SUP./MFR.: Viscon		DRAWING NO.: PDSB UPG1818 RIW	REV.: 00
REV.	DATE	DRN/CHK/REV	REVISED	AZMAN			SUBSTATION: PENCAWANG ELEKTRIK 11kV		THIS SHEET: 3-8-1A NEXT SHEET: 3-8-2



EXISTING DIGITAL INPUT MODULE WIRING DIAGRAM (RMU TYPE) - 2

ENGINEERING SERVICES DISEMAK 21 SEP 2023

GURSHARAN SINGH 10.9.23 ES DISTRIBUTION NETWORK DIVISION - TNB

			DATE	17.07.23	CLIENT: TENAGA NASIONAL BERHAD	CONTRACTOR: Viscon	TITLE: TELECONTROL OVERVIEW DRAWING	= PE	+ RTU /
			DRAWN	KHAIRI			DIGITAL INPUT MODULE WIRING DIAGRAM - 2		
			CHECKED	NAZMI	CONTRACT NO: TNB 1799/2023	SUP./MFR.: Viscon		DRAWING NO.: PDSB UPG1818 RIW	REV.: 00
REV.	DATE	DRN/CHK/REV	REVISED	AZMAN			SUBSTATION: PENCAWANG ELEKTRIK 11kV		THIS SHEET: 3-8-2 / NEXT SHEET: 3-8-2A

INTERNAL I/O CABLE IN CABLE DUCT

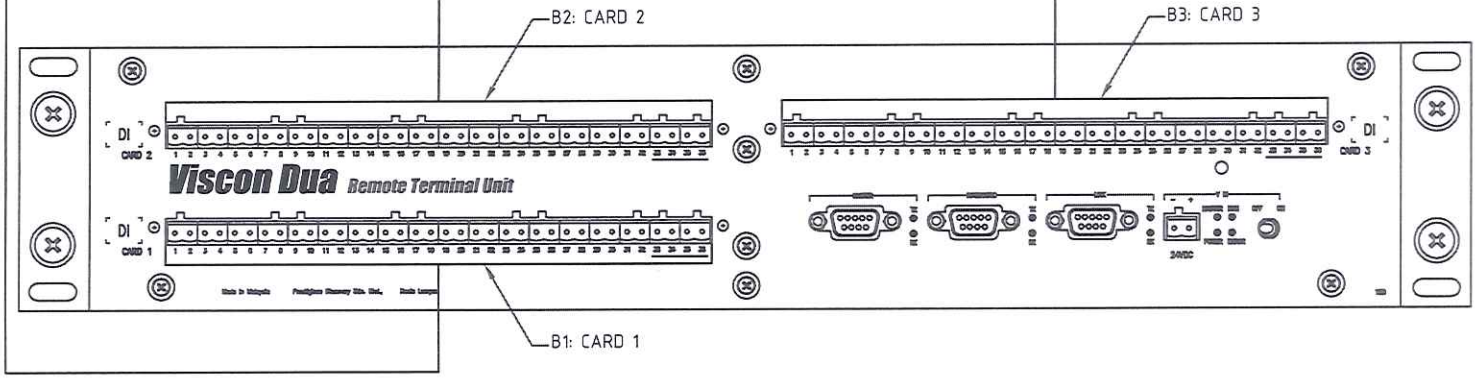
LINKED TO CARD 1

INTERNAL I/O CABLE IN CABLE DUCT

LINKED TO CARD 2

INTERNAL I/O CABLE IN CABLE DUCT

LINKED TO CARD 3



I/O POINTS

1
2
31
32
33 COMM
34 COMM
35 COMM
36 COMM

I/O POINTS

1
2
31
32
33 COMM
34 COMM
35 COMM
36 COMM

I/O POINTS

1
2
31
32
33 COMM
34 COMM
35 COMM
36 COMM

COMMON FROM PLANT

37
38
39
40

+VS
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7
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11
13

-VS
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8
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14

ENGINEERING SERVICES DISEMAK 21 SEP 2023 DISTRIBUTION NETWORK DIVISION

GURSHARAN SINGH 10 AUG 23 ES DISTRIBUTION NETWORK DIVISION TNB

EXISTING DIGITAL INPUT MODULE WIRING DIAGRAM (VCB TYPE) - 2

			DATE	17.07.23	CLIENT: TENAGA NASIONAL BERHAD	CONTRACTOR: Viscon	TITLE: TELECONTROL OVERVIEW DRAWING	= PE	+ RTU /
			DRAWN	KHAIRI			DIGITAL INPUT MODULE WIRING DIAGRAM - 2		
			CHECKED	NAZMI	CONTRACT NO: TNB 1799/2023	SUP./MFR.: Viscon		DRAWING NO.: PDSB UPG1818 RIW	REV.: 00
REV.	DATE	DRN/CHK/REV	REVISED	AZMAN			SUBSTATION: PENCAWANG ELEKTRIK 11kV		THIS SHEET: 3-8-2A NEXT SHEET: 3-9-1

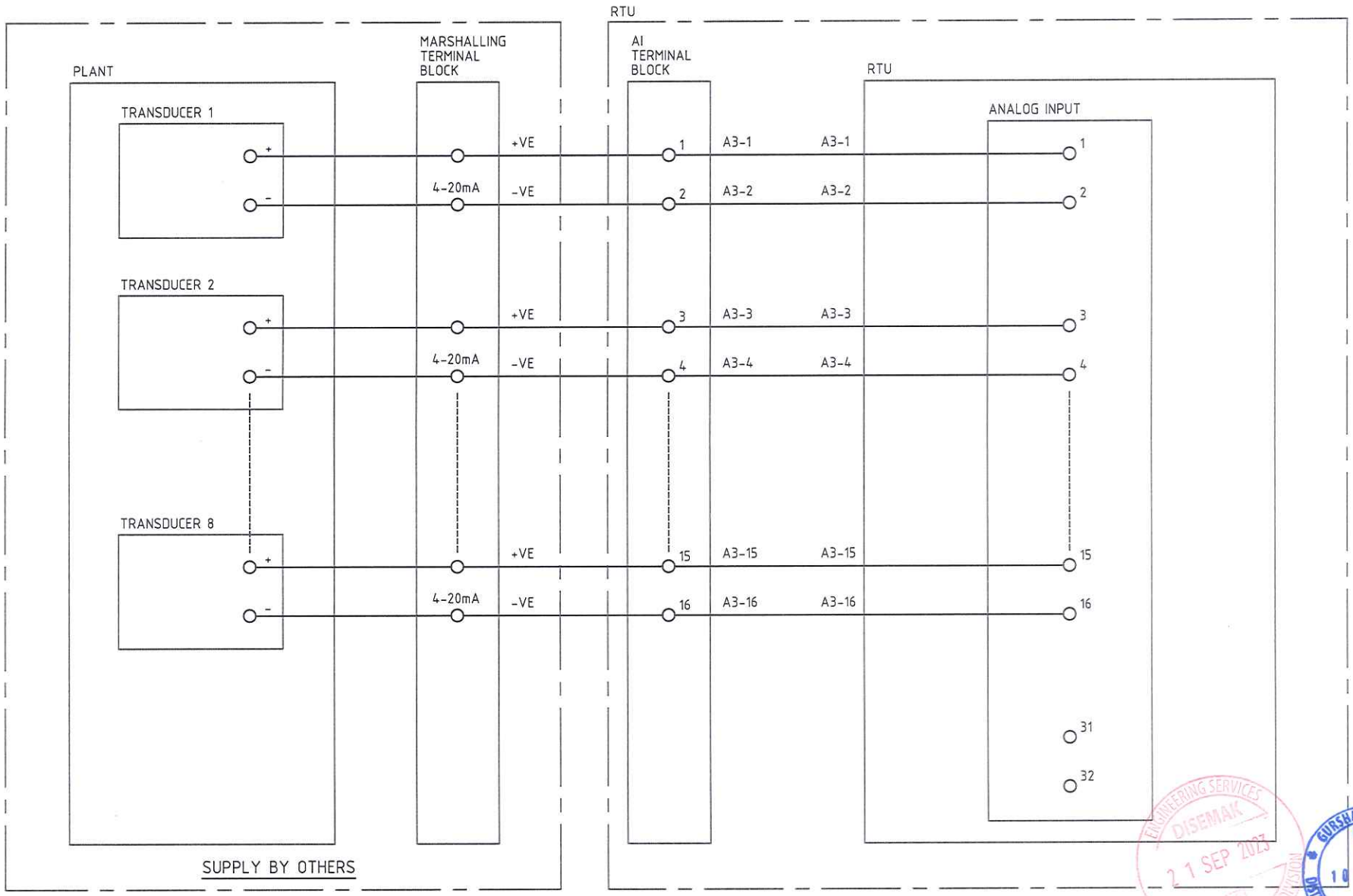
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EXISTING ANALOG INPUT MODULE WIRING DIAGRAM (RMU TYPE) - 1

ENGINEERING SERVICES DISEMAK 21 SEP 2023 TNB DISTRIBUTION NETWORK DIVISION

GURSHARAN SINGH 10 AUG 23 ES DISTRIBUTION NETWORK DIVISION, TNB

			DATE	17.07.23	CLIENT: TENAGA NASIONAL BERHAD	CONTRACTOR: Viscon	TITLE: TELECONTROL OVERVIEW DRAWING	= PE	+ RTU /
			DRAWN	KHAIRI					
			CHECKED	NAZMI	CONTRACT NO: TNB 1799/2023	SUP./MFR.: Viscon	ANALOG INPUT MODULE WIRING DIAGRAM - 1	DRAWING NO.: PDSB UPG1818 RIW	REV.: 00
REV.	DATE	DRN/CHK/REV	REVISED	AZMAN			SUBSTATION: PENCAWANG ELEKTRIK 11kV		THIS SHEET: 3-9-1 / NEXT SHEET: 3-9-1A

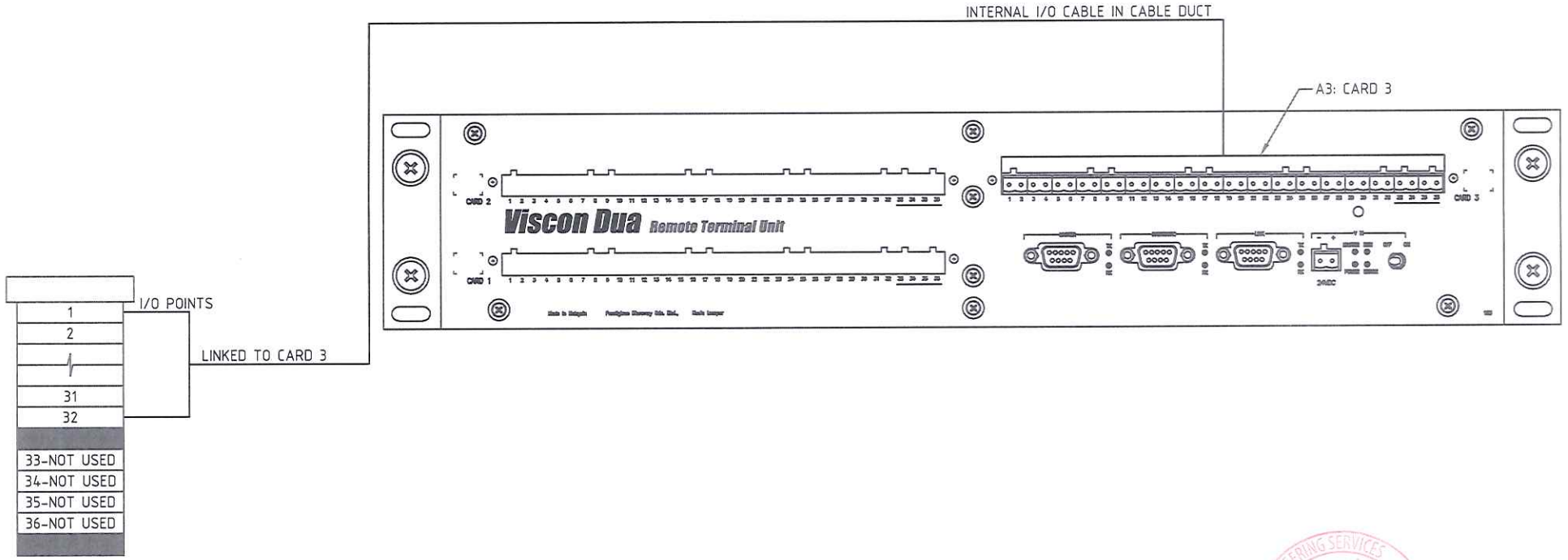


EXISTING ANALOG INPUT MODULE WIRING DIAGRAM (VCB TYPE) - 1

ENGINEERING SERVICES DISEMAK 21 SEP 2023 TNB NETWORK DIVISION

GURSHARAN SINGH 10 AUG 23 ES DISTRIBUTION NETWORK DIVISION TNB

			DATE	17.07.23	CLIENT: TENAGA NASIONAL BERHAD	CONTRACTOR: Viscon	TITLE: TELECONTROL OVERVIEW DRAWING	= PE	+ RTU	/
			DRAWN	KHAIRI						
			CHECKED	NAZMI	CONTRACT NO: TNB 1799/2023	SUP./MFR.: Viscon	ANALOG INPUT MODULE WIRING DIAGRAM - 1	DRAWING NO.: PDSB UPG1818 RIW	REV.: 00	THIS SHEET: 3-9-1A
REV.	DATE	DRN/CHK/REV	REVISED	AZMAN			SUBSTATION: PENCAWANG ELEKTRIK 11kV			NEXT SHEET: 3-9-2

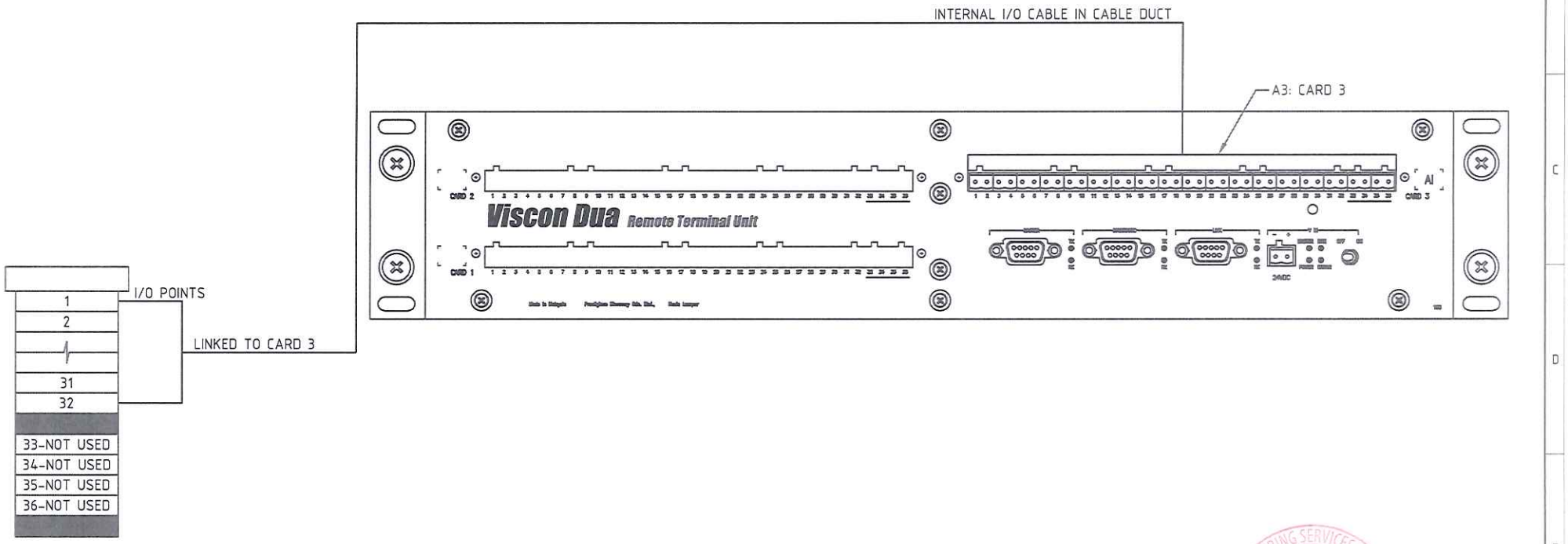


EXISTING ANALOG INPUT MODULE WIRING DIAGRAM - 2

ENGINEERING SERVICES DISEMAK 21 SEP 2023 DISTRIBUTION NETWORK DIVISION

GURSHARAN SINGH 10 AUG 23 ES DISTRIBUTION NETWORK DIVISION, TNB

			DATE	17.07.23	CLIENT: TENAGA NASIONAL BERHAD	CONTRACTOR: Viscon	TITLE: TELECONTROL OVERVIEW DRAWING	= PE	+ RTU /
			DRAWN	KHAIRI			ANALOG INPUT MODULE WIRING DIAGRAM - 2	DRAWING NO.: PDSB UPG1818 RIW	REV.: 00
			CHECKED	NAZMI	CONTRACT NO: TNB 1799/2023	SUP./MFR.: Viscon	SUBSTATION: PENCAWANG ELEKTRIK 11kV		THIS SHEET: 3-9-2
REV.	DATE	DRN/CHK/REV	REVISED	AZMAN					NEXT SHEET: 3-9-2A



EXISTING ANALOG INPUT MODULE WIRING DIAGRAM (VCB TYPE) - 2

ENGINEERING SERVICES DISEMAK 21 SEP 2023 TNB DISTRIBUTION NETWORK DIVISION

GURSHARAN SINGH 10 AUG 23 ES DISTRIBUTION NETWORK DIVISION, TNB

REV.	DATE	DRN/CHK/REV		
			DATE	17.07.23
			DRAWN	KHAIRI
			CHECKED	NAZMI
			REVISED	AZMAN

CLIENT: TENAGA NASIONAL BERHAD

CONTRACT NO: TNB 1799/2023

CONTRACTOR: Viscon

SUP./MFR.: Viscon

TITLE: TELECONTROL OVERVIEW DRAWING

ANALOG INPUT MODULE WIRING DIAGRAM - 2

SUBSTATION: PENCAWANG ELEKTRIK 11kV

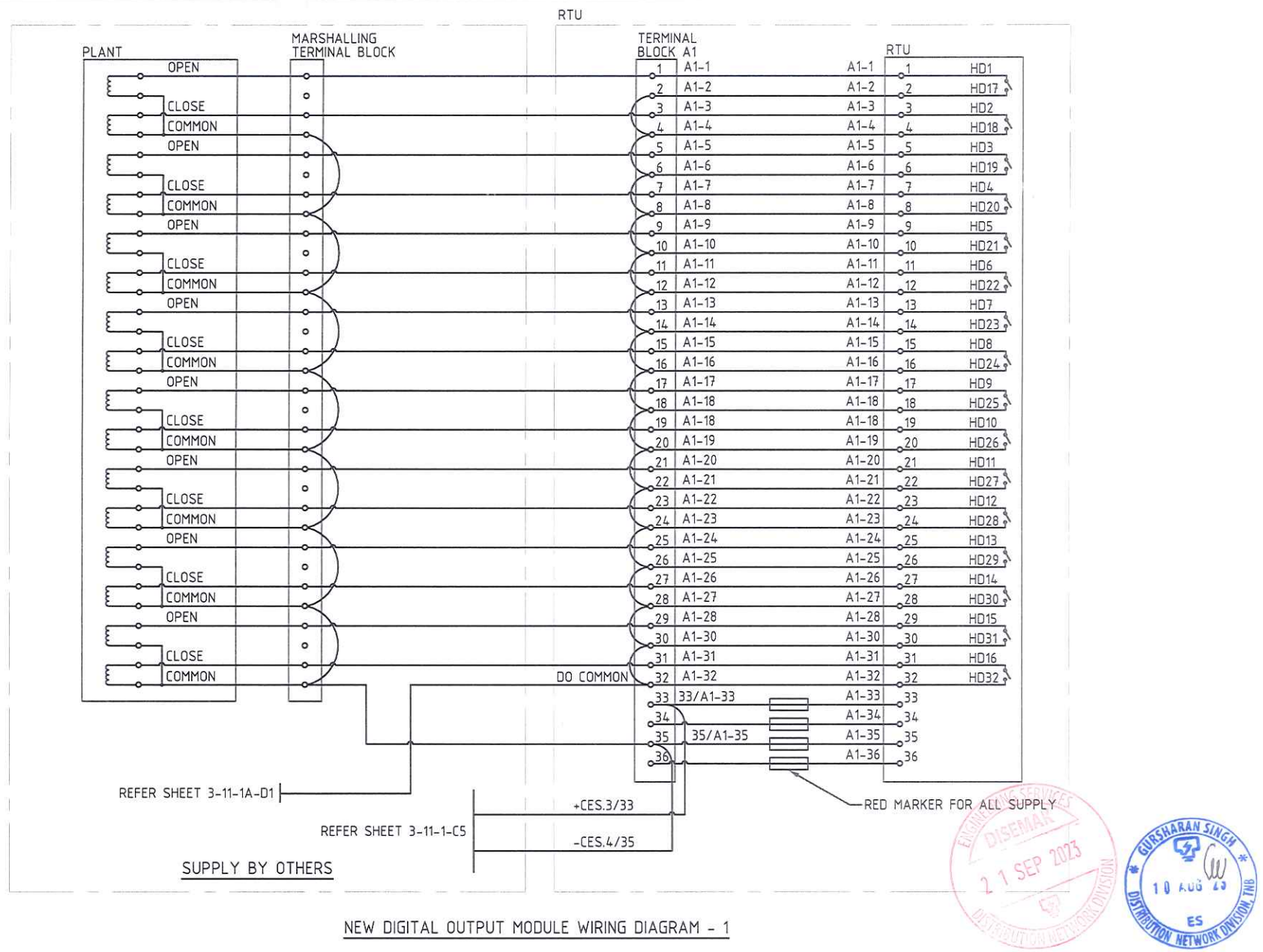
= PE | + RTU | /

DRAWING NO.: PDSB UPG1818 RIW

REV.: 00

THIS SHEET: 3-9-2A

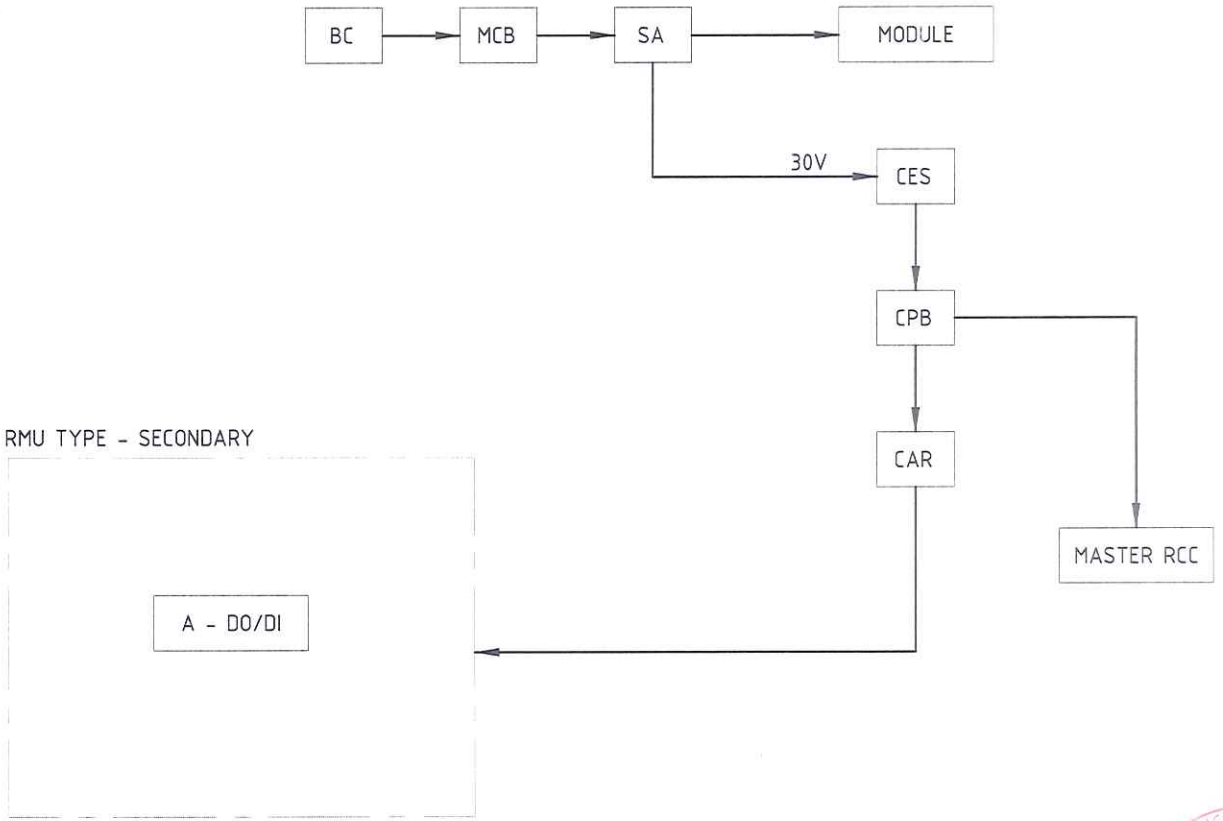
NEXT SHEET: 3-10-1



NEW DIGITAL OUTPUT MODULE WIRING DIAGRAM - 1

			DATE	17.07.23	CLIENT: TENAGA NASIONAL BERHAD	CONTRACTOR: Viscon	TITLE: TELECONTROL OVERVIEW DRAWING	= PE	+ RTU	/	
			DRAWN	KHAIRI			DIGITAL OUTPUT MODULE WIRING DIAGRAM - 1	DRAWING NO.: PDSB UPG1818 RIW	REV.: 00	THIS SHEET: 3-10-1	
			CHECKED	NAZMI	CONTRACT NO: TNB 1799/2023	SUP./MFR.: Viscon	SUBSTATION: PENCAWANG ELEKTRIK 11kV			NEXT SHEET: 3-10-1A	
REV.	DATE	DRN/CHK/REV	REVISED	AZMAN							

ENGINEERING SERVICES DISEMAK 21 SEP 2023 DISTRIBUTION NETWORK DIVISION
GURSHARAN SINGH 10 AUG 23 ES DISTRIBUTION NETWORK DIVISION TNB



(30V) (RMU TYPE) - CONTROL FLOW DIAGRAM

ENGINEERING SERVICES DISEMAK 21 SEP 2023 TNB DISTRIBUTION DIVISION

GURSHARAN SINGH 10 AUG 23 ES DISTRIBUTION NETWORK DIVISION, TNB

REV.	DATE	DRN/CHK/REV		
			DATE	17.07.23
			DRAWN	KHAIRI
			CHECKED	NAZMI
			REVISED	AZMAN

CLIENT: TENAGA NASIONAL BERHAD

CONTRACT NO: TNB 1799/2023

CONTRACTOR: Viscon

SUP./MFR.: Viscon

TITLE: TELECONTROL OVERVIEW DRAWING

30V (RMU TYPE) - CONTROL FLOW DIAGRAM

SUBSTATION: PENCAWANG ELEKTRIK 11kV

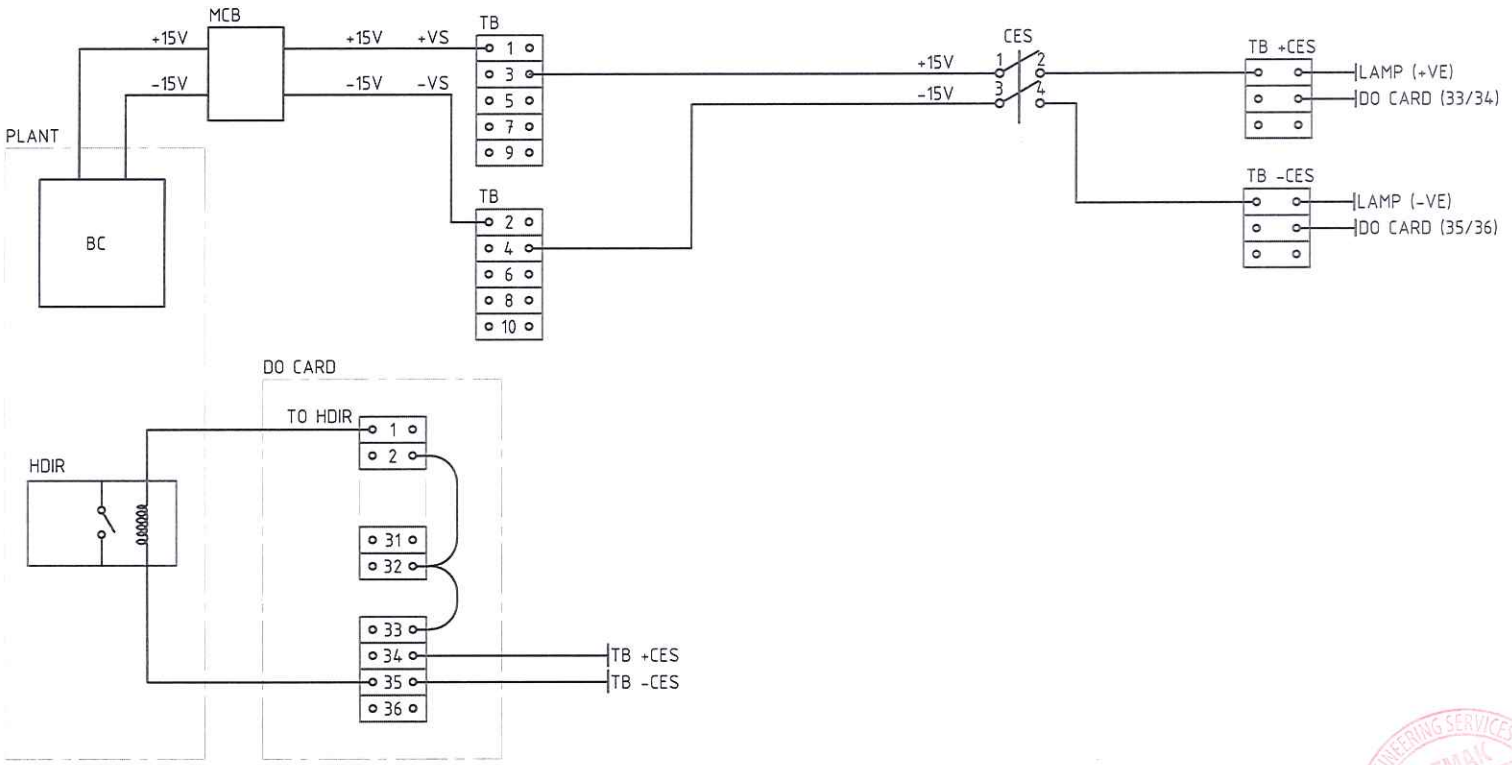
= PE | + RTU | /

DRAWING NO.: PDSB UPG1818 RIW

REV.: 00

THIS SHEET: 3-10-1A

NEXT SHEET: 3-11-1B



(30V) (RMU TYPE) - BEFORE MODIFICATION WIRING DIAGRAM

ENGINEERING SERVICES DISEMAK 21 SEP 2023 DISTRIBUTION NETWORK DIVISION

GURSHARAN SINGH 10 AUG 23 ES DISTRIBUTION NETWORK DIVISION, TNB

REV.	DATE	DRN/CHK/REV		
			DATE	17.07.23
			DRAWN	KHAIRI
			CHECKED	NAZMI
			REVISED	AZMAN

CLIENT: TENAGA NASIONAL BERHAD

CONTRACT NO: TNB 1799/2023

CONTRACTOR: Viscon

SUP./MFR.: Viscon

TITLE: TELECONTROL OVERVIEW DRAWING

30V (RMU) - BEFORE MODIFICATION WIRING DIAGRAM

SUBSTATION: PENCAWANG ELEKTRIK 11kV

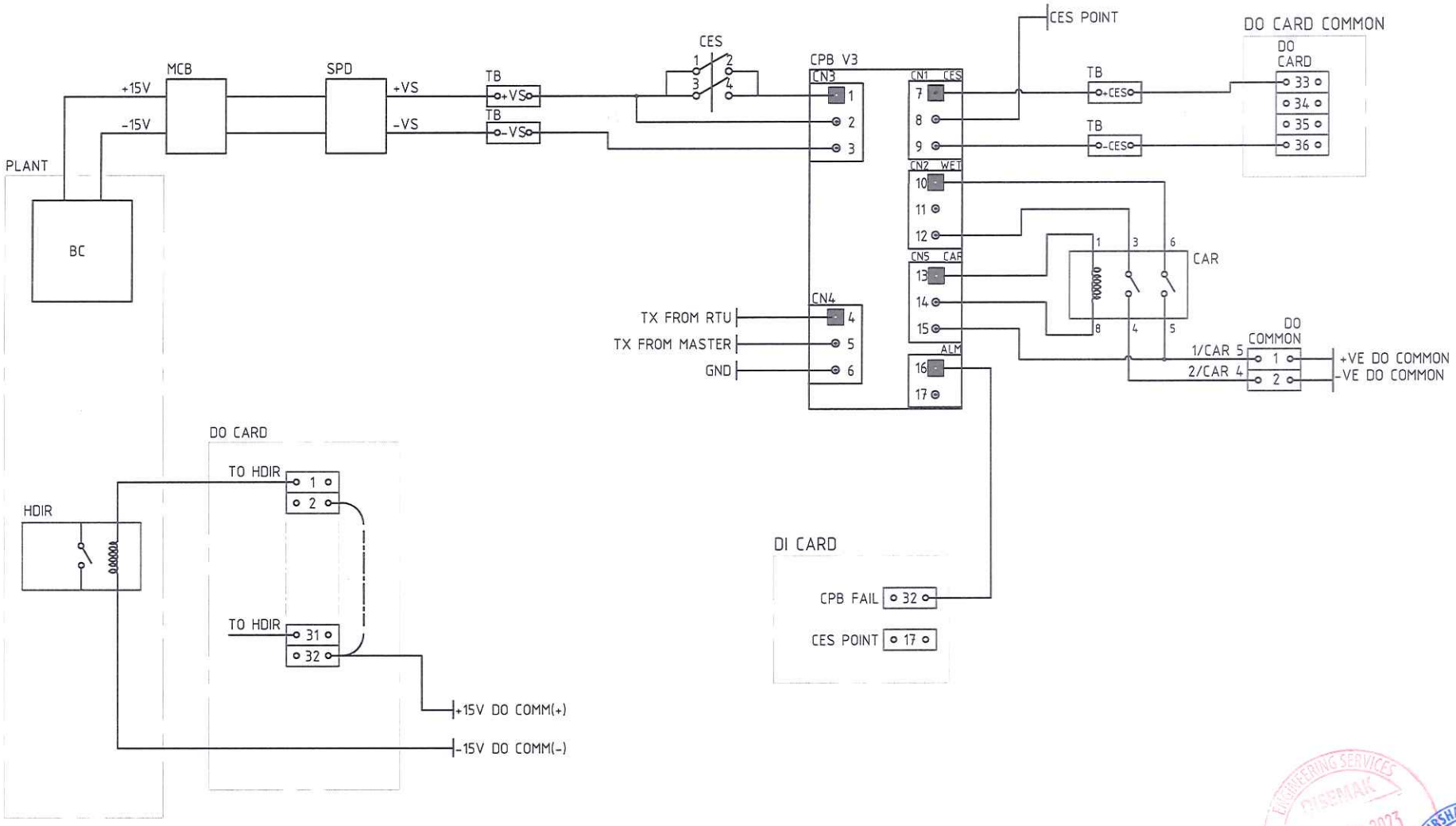
= PE | + RTU | /

DRAWING NO.: PDSB UPG1818 RIW

REV.: 00

THIS SHEET: 3-10-1B

NEXT SHEET: 3-11-1C

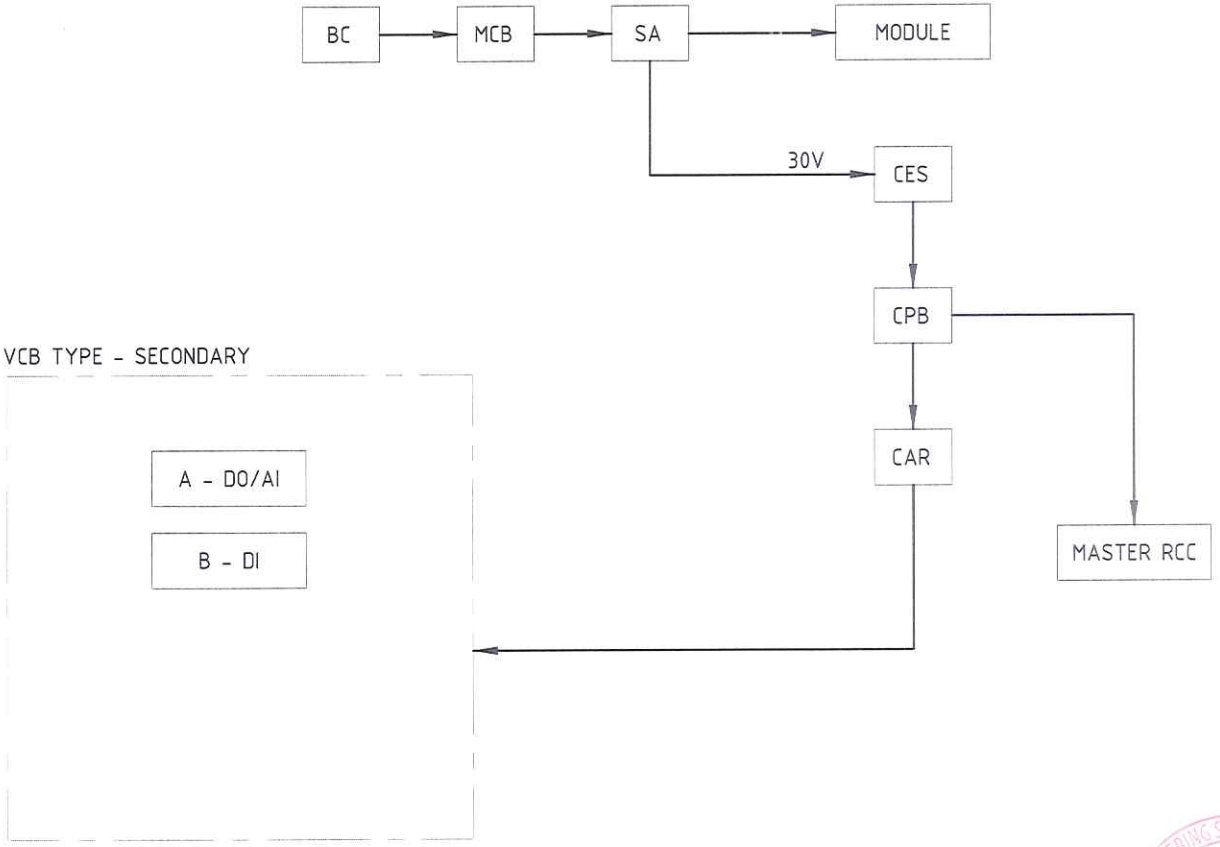


(30V) (RMU TYPE) - AFTER MODIFICATION WIRING DIAGRAM

ENGINEERING SERVICES DISSEMAK 21 SEP 2023

GURSHARAN SINGH 10 AUG 23 ES DISTRIBUTION NETWORK DIVISION, TNB

			DATE	17.07.23	CLIENT: TENAGA NASIONAL BERHAD	CONTRACTOR: Viscon	TITLE: TELECONTROL OVERVIEW DRAWING	= PE	+ RTU	/
			DRAWN	KHAIRI			30V (RMU) - AFTER MODIFICATION WIRING DIAGRAM	DRAWING NO.: PDSB UPG1818 RIW	REV.: 00	THIS SHEET: 3-10-1C
			CHECKED	NAZMI	CONTRACT NO: TNB 1799/2023	SUP./MFR.: Viscon	SUBSTATION: PENCAWANG ELEKTRIK 11kV			NEXT SHEET: 3-10-1D
REV.	DATE	DRN/CHK/REV	REVISED	AZMAN						

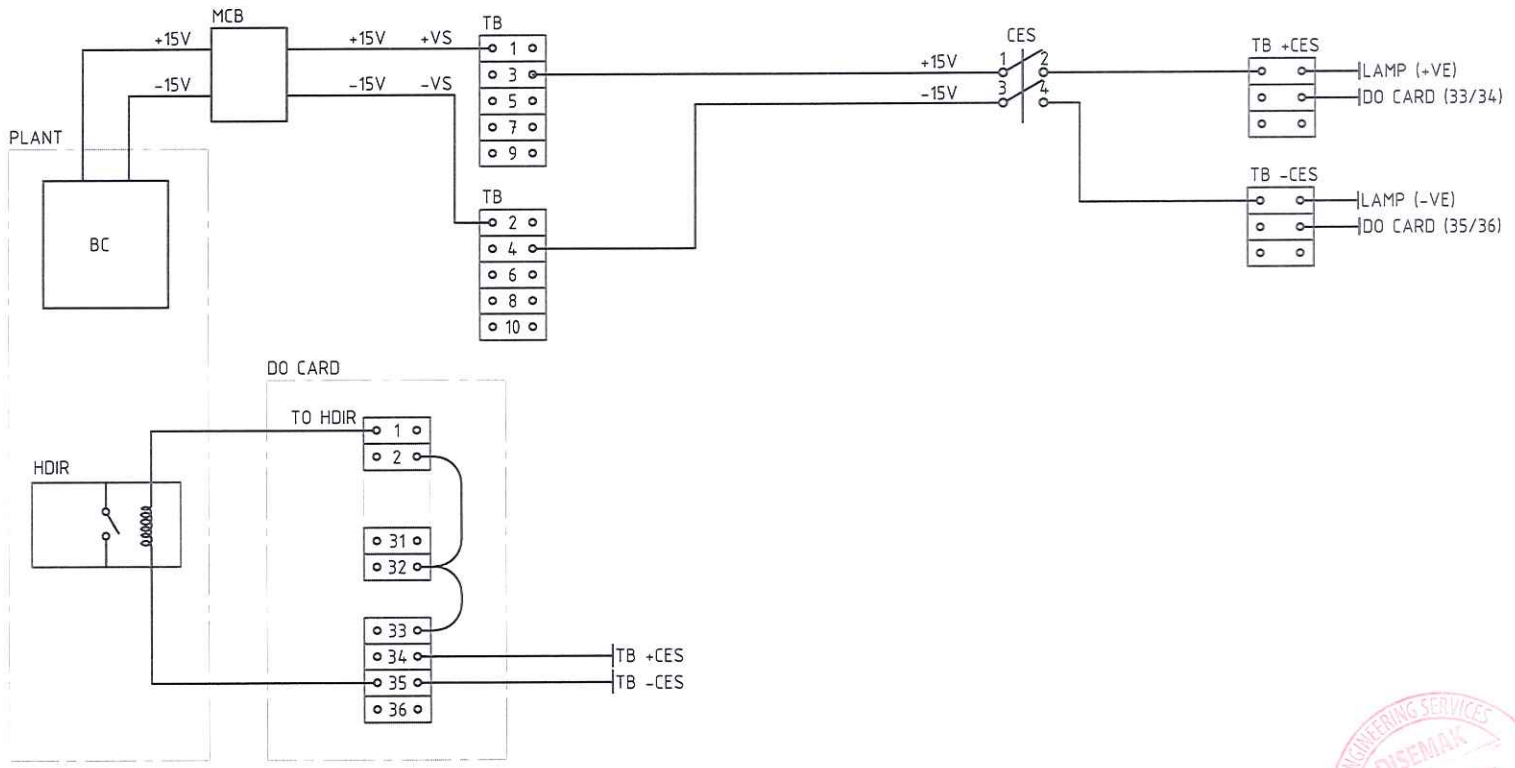


30V (VCB TYPE) - CONTROL FLOW DIAGRAM

ENGINEERING SERVICES DISEMAK 21 SEP 2023 DISTRIBUTION NETWORK

GURSHARAN SINGH 10 Aug 23 ES DISTRIBUTION NETWORK DIVISION TNB

			DATE	17.07.23	CLIENT: TENAGA NASIONAL BERHAD	CONTRACTOR: Viscon	TITLE: TELECONTROL OVERVIEW DRAWING	= PE	+ RTU /
			DRAWN	KHAIRI			30V (VCB TYPE) - CONTROL FLOW DIAGRAM	DRAWING NO.: PDSB UPG1818 RIW	REV.: 00
			CHECKED	NAZMI	CONTRACT NO: TNB 1799/2023	SUP./MFR.: Viscon	SUBSTATION: PENCAWANG ELEKTRIK 11kV		THIS SHEET: 3-10-1D
REV.	DATE	DRN/CHK/REV	REVISED	AZMAN					NEXT SHEET: 3-10-1E

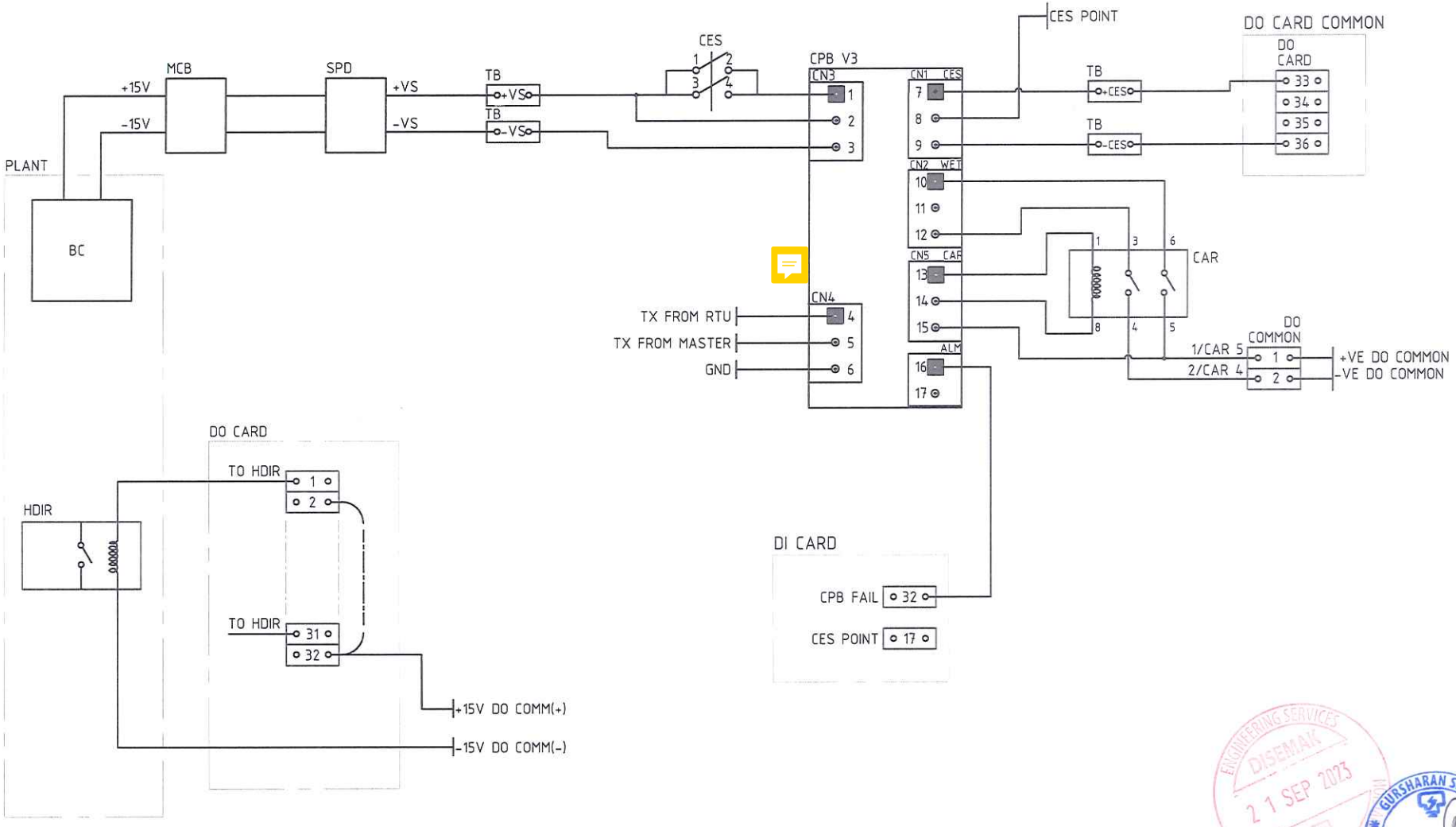


30V (VCB TYPE) - BEFORE MODIFICATION WIRING DIAGRAM

ENGINEERING SERVICES DISEMAK 21 SEP 2023 DISTRIBUTION NETWORK MANAGEMENT

GURSHARAN SINGH 10 ES DISTRIBUTION NETWORK MANAGEMENT

			DATE	17.07.23	CLIENT: TENAGA NASIONAL BERHAD	CONTRACTOR: Viscon	TITLE: TELECONTROL OVERVIEW DRAWING	= PE	+ RTU /
			DRAWN	KHAIRI			30V (VCBTYPE) - BEFORE MODIFICATION WIRING DIAGRAM	DRAWING NO.: PDSB UPG1818 RIW	REV.: 00
			CHECKED	NAZMI	CONTRACT NO: TNB 1799/2023	SUP./MFR.: Viscon	SUBSTATION: PENCAWANG ELEKTRIK 11kV		THIS SHEET: 3-10-1E
REV.	DATE	DRN/CHK/REV	REVISED	AZMAN					NEXT SHEET: 3-10-1F

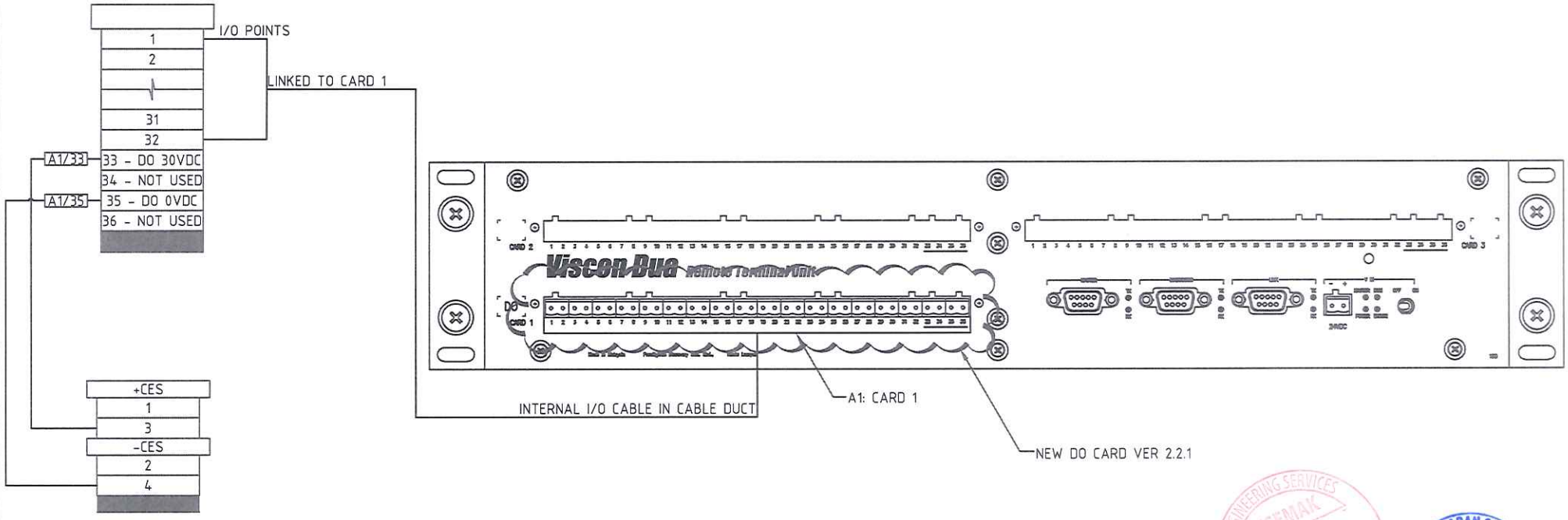


30V (VCB TYPE) - AFTER MODIFICATION WIRING DIAGRAM

ENGINEERING SERVICES DISEMAK 21 SEP 2023 DISTRIBUTION NETWORK DIVISION TNB

GURSHARAN SINGH 10 AUG 23 ES DISTRIBUTION NETWORK DIVISION TNB

			DATE	17.07.23	CLIENT: TENAGA NASIONAL BERHAD	CONTRACTOR: Viscon	TITLE: TELECONTROL OVERVIEW DRAWING	= PE	+ RTU /
			DRAWN	KHAIRI			30V (VCB) - AFTER MODIFICATION WIRING DIAGRAM		
			CHECKED	NAZMI	CONTRACT NO: TNB 1799/2023	SUP./MFR.: Viscon		DRAWING NO.: PDSB UPG1818 RIW	REV.: 00
REV.	DATE	DRN/CHK/REV	REVISED	AZMAN			SUBSTATION: PENCAWANG ELEKTRIK 11kV		THIS SHEET: 3-10-1F NEXT SHEET: 3-10-2

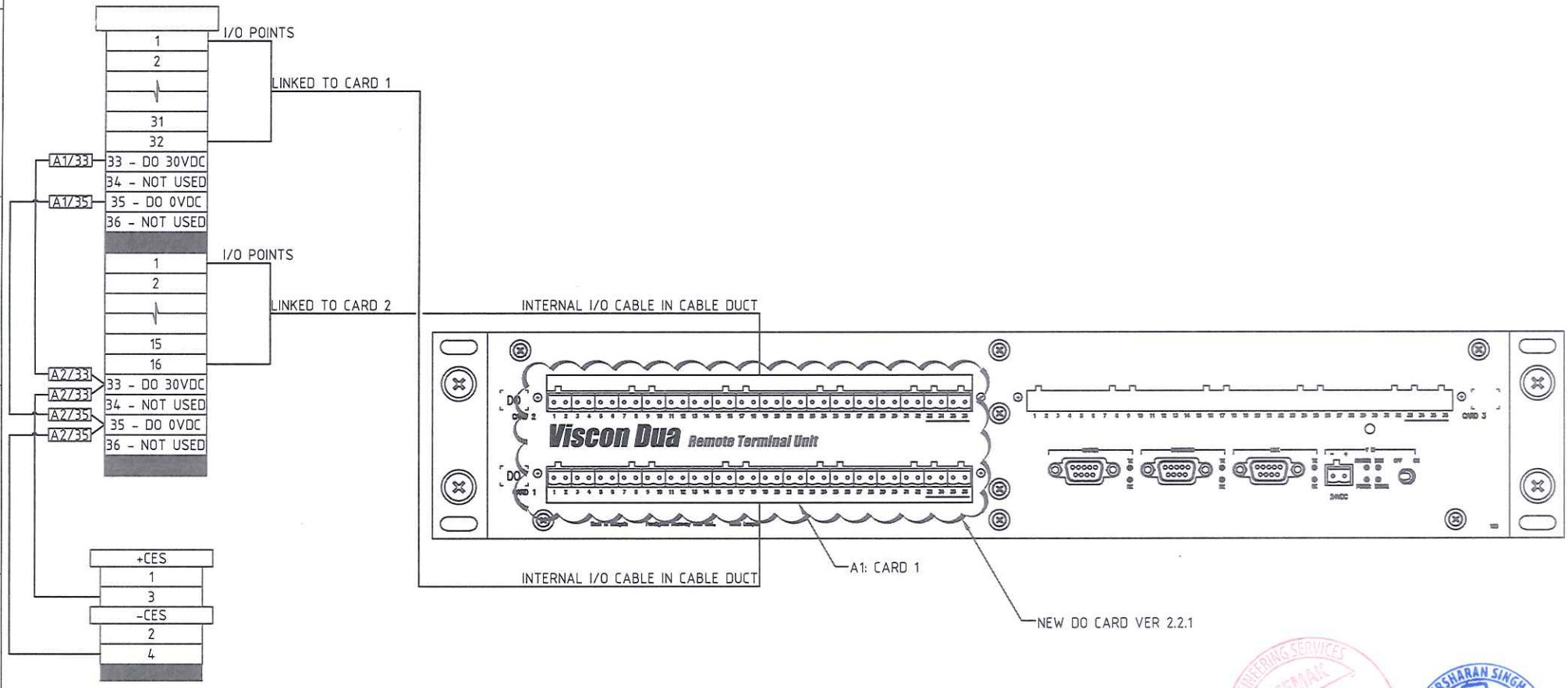


NEW DIGITAL OUTPUT MODULE WIRING DIAGRAM (RMU TYPE) - 2

ENGINEERING SERVICES DISEMAK 21 SEP 2023 DISTRIBUTION NETWORK DIVISION

GURSHARAN SINGH 10 AUG 23 ES DISTRIBUTION NETWORK DIVISION TNB

			DATE	17.07.23	CLIENT: TENAGA NASIONAL BERHAD	CONTRACTOR: Viscon	TITLE: TELECONTROL OVERVIEW DRAWING	= PE	+ RTU /
			DRAWN	KHAIRI			DIGITAL OUTPUT MODULE WIRING DIAGRAM - 2	DRAWING NO.: PDSB UPG1818 RIW	REV.: 00
			CHECKED	NAZMI	CONTRACT NO: TNB 1799/2023	SUP./MFR.: Viscon	SUBSTATION: PENCAWANG ELEKTRIK 11kV		THIS SHEET: 3-10-2
REV.	DATE	DRN/CHK/REV	REVISED	AZMAN					NEXT SHEET: 3-10-2A

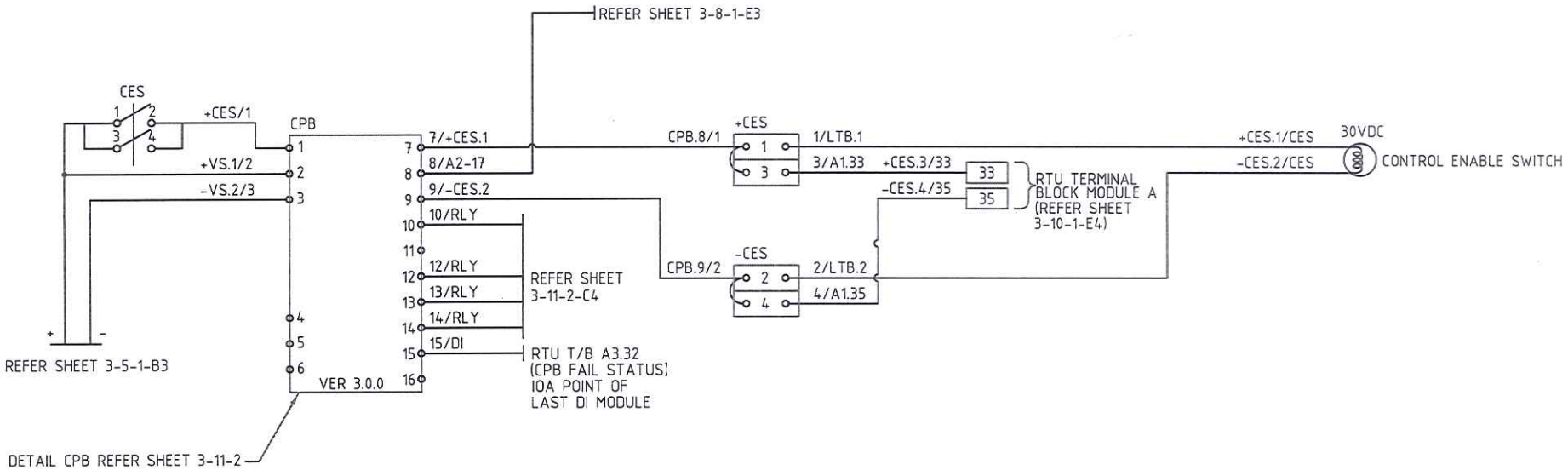


NEW DIGITAL OUTPUT MODULE WIRING DIAGRAM (VCB TYPE) - 2

ENGINEERING SERVICES DISEMAK 21 SEP 2023 DISTRIBUTION NETWORK DIVISION

GURSHARAN SINGH 10 AUG 23 ES DISTRIBUTION NETWORK DIVISION, TNB

			DATE	17.07.23	CLIENT: TENAGA NASIONAL BERHAD	CONTRACTOR: Viscon	TITLE: TELECONTROL OVERVIEW DRAWING	= PE	+ RTU /
			DRAWN	KHAIRI			DIGITAL OUTPUT MODULE WIRING DIAGRAM - 2		
			CHECKED	NAZMI	CONTRACT NO: TNB 1799/2023	SUP./MFR.: Viscon		DRAWING NO.: PDSB UPG1818 RIW	REV.: 00
REV.	DATE	DRN/CHK/REV	REVISED	AZMAN			SUBSTATION: PENCAWANG ELEKTRIK 11kV		THIS SHEET: 3-10-2A NEXT SHEET: 3-11-1

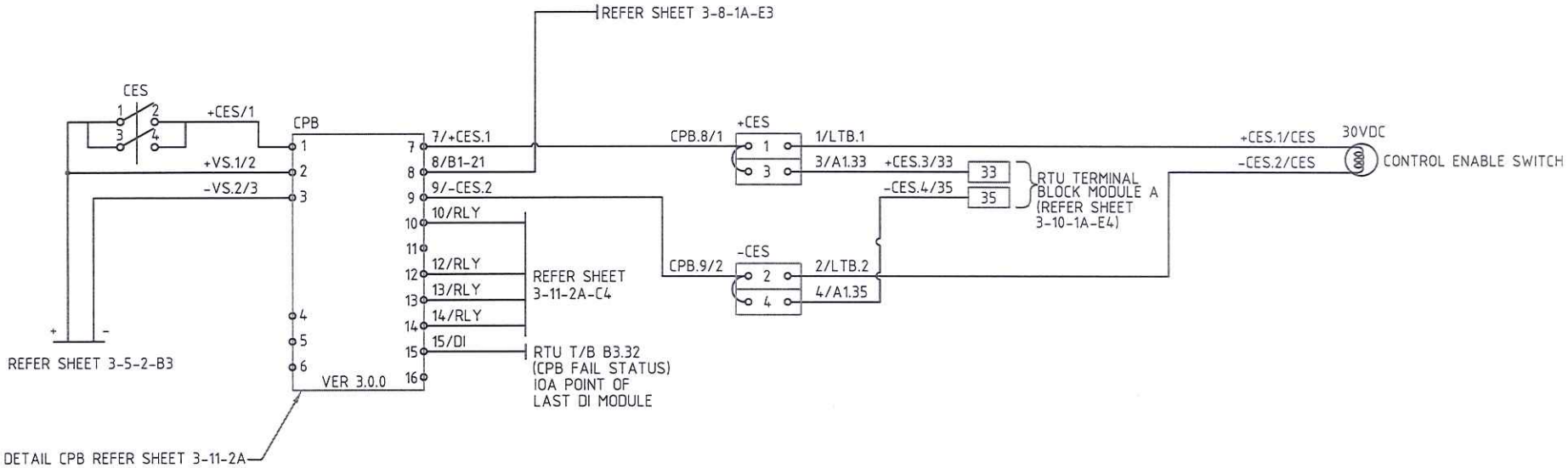


NEW CONTROL ENABLE SWITCH CIRCUIT DIAGRAM (RMU TYPE)

ENGINEERING SERVICES DISEMAK 21 SEP 2023

GURSHARAN SINGH 14 SEP 23 ES DISTRIBUTION NETWORK DIVISION, TNB

			DATE	17.07.23	CLIENT: TENAGA NASIONAL BERHAD	CONTRACTOR: Viscon	TITLE: TELECONTROL OVERVIEW DRAWING	= PE	+ RTU	/
			DRAWN	KHAIRI			CONTROL ENABLE SWITCH CIRCUIT DIAGRAM			
			CHECKED	NAZMI	CONTRACT NO: TNB 1799/2023	SUP./MFR.: Viscon		DRAWING NO.: PDSB UPG1818 RIW	REV.: 00	THIS SHEET: 3-11-1
REV.	DATE	DRN/CHK/REV	REVISED	AZMAN			SUBSTATION: PENCAWANG ELEKTRIK 11kV			NEXT SHEET: 3-11-1A

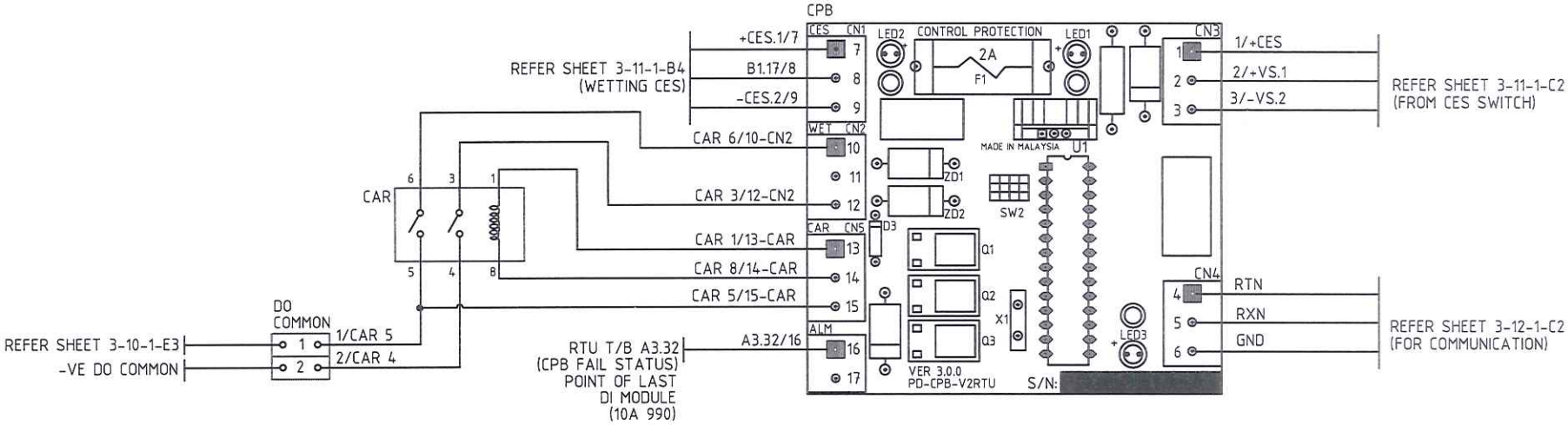


NEW CONTROL ENABLE SWITCH CIRCUIT DIAGRAM (VCB TYPE)

ENGINEERING SERVICES DISMAK 21 SEP 2023

GURSHARAN SINGH 14 SE... DISTRIBUTION NETWORK DIVISION

			DATE	17.07.23	CLIENT: TENAGA NASIONAL BERHAD	CONTRACTOR: Viscon	TITLE: TELECONTROL OVERVIEW DRAWING	= PE	+ RTU	/
			DRAWN	KHAIRI			CONTROL ENABLE SWITCH CIRCUIT DIAGRAM			
			CHECKED	NAZMI	CONTRACT NO: TNB 1799/2023	SUP./MFR.: Viscon		DRAWING NO.: PDSB UPG1818 RIW	REV.: 00	THIS SHEET: 3-11-1A
REV.	DATE	DRN/CHK/REV	REVISED	AZMAN			SUBSTATION: PENCAWANG ELEKTRIK 11kV			NEXT SHEET: 3-11-2



NEW CPB WIRING CIRCUIT (RMU TYPE)

NOTE:

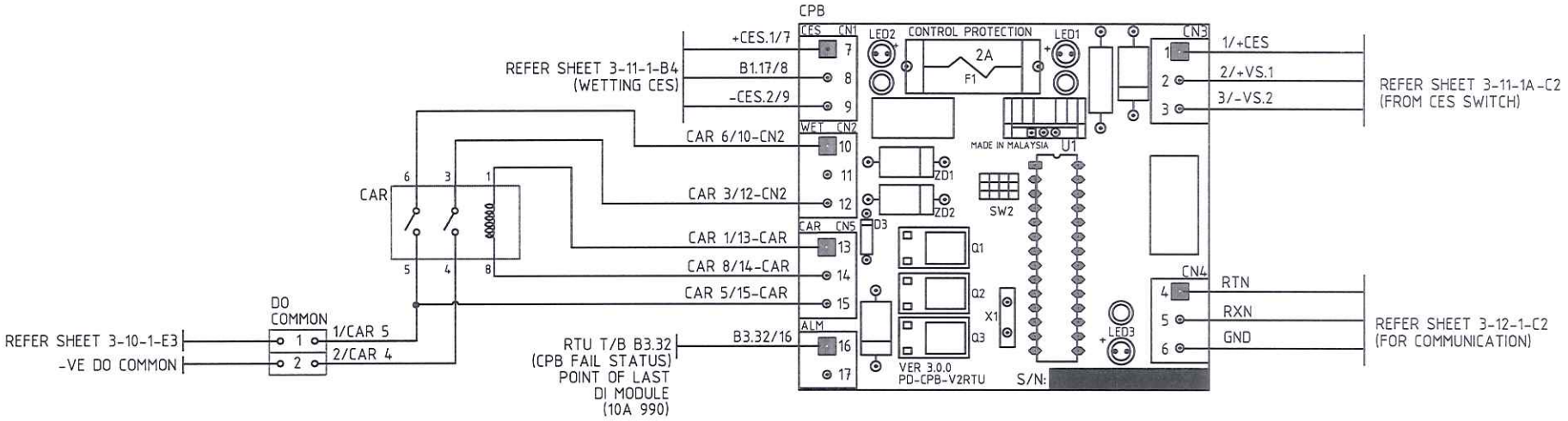
NO.	DESCRIPTION	ACTION
1.	COMM FAIL / DISCONNECT FROM RCC	VOLTAGE FOR 'CHECK CES' WILL CUT OFF AND CPB WILL TRIGGER 'CPB ALARM'.
2.	FUSE BLOW	CPB WILL TRIGGER 'CPB ALARM'.
3.	CPB RELAY FAIL	VOLTAGE FOR WETTING DO CARD WILL CUT OFF AND CPB WILL TRIGGER 'CPB ALARM'.

ENGINEERING SERVICES DISEMAK 21 SEP 2023

GURSHARAN SINGH 14 SEP '23 ES DISTRIBUTION NETWORK DIVISION, TNB

REV.	DATE	DRN/CHK/REV		
			DATE	17.07.23
			DRAWN	KHAIRI
			CHECKED	NAZMI
			REVISED	AZMAN

CLIENT: TENAGA NASIONAL BERHAD
CONTRACT NO: TNB 1799/2023
CONTRACTOR: Viscon
SUP./MFR.: Viscon
TITLE: TELECONTROL OVERVIEW DRAWING
CPB WIRING CIRCUIT
SUBSTATION: PENCAWANG ELEKTRIK 11kV
= PE + RTU /
DRAWING NO.: PDSB UPG1818 RIW
REV.: 00
THIS SHEET: 3-11-2
NEXT SHEET: 3-11-2A



NEW CPB WIRING CIRCUIT (VCB TYPE)

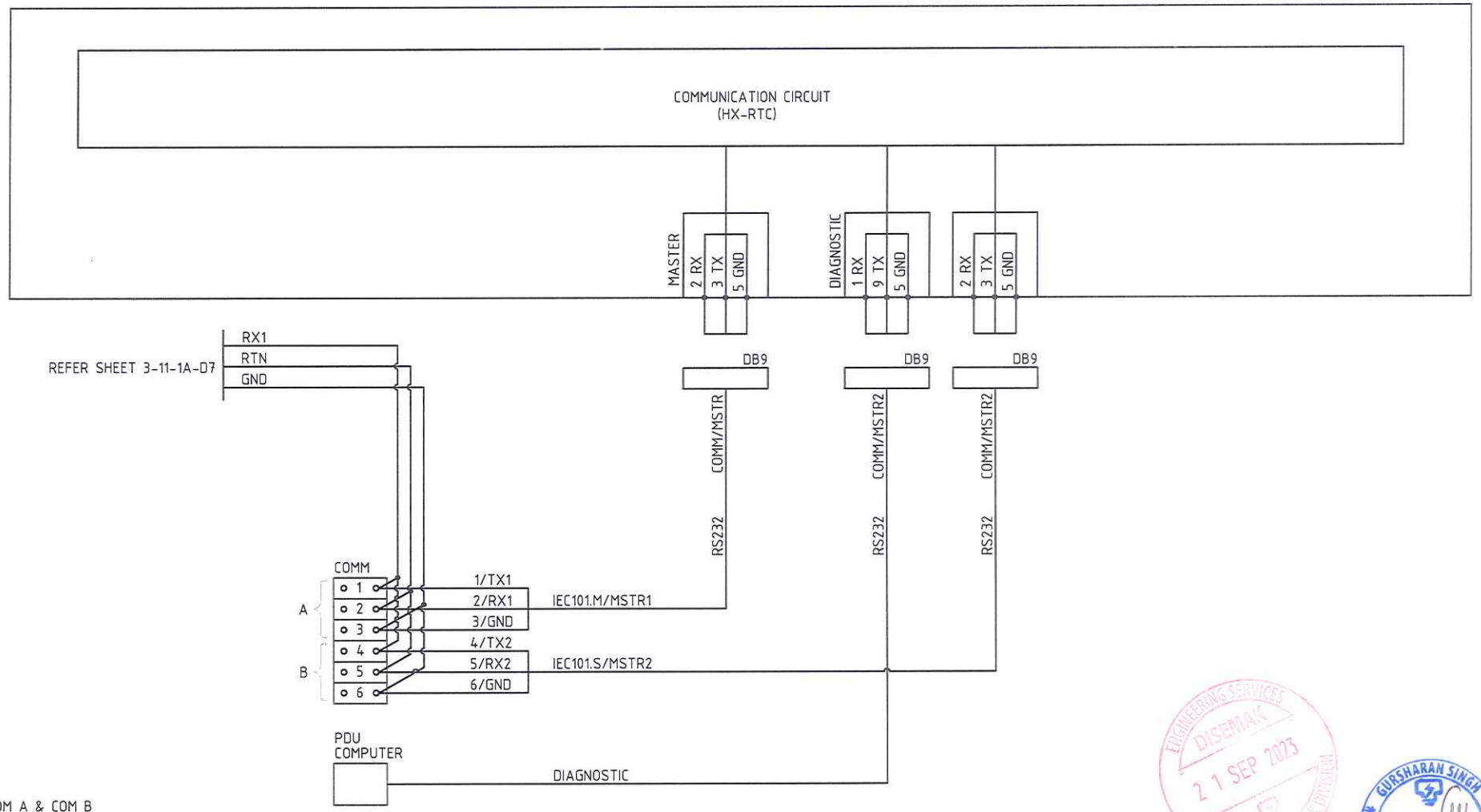
NOTE:

NO.	DESCRIPTION	ACTION
1.	COMM FAIL / DISCONNECT FROM RCC	VOLTAGE FOR 'CHECK CES' WILL CUT OFF AND CPB WILL TRIGGER 'CPB ALARM'.
2.	FUSE BLOW	CPB WILL TRIGGER 'CPB ALARM'.
3.	CPB RELAY FAIL	VOLTAGE FOR WETTING DO CARD WILL CUT OFF AND CPB WILL TRIGGER 'CPB ALARM'.

ENGINEERING SERVICES DISEMAK 21 SEP 2023

REV.	DATE	DRN/CHK/REV		
			DATE	17.07.23
			DRAWN	KHAIRI
			CHECKED	NAZMI
			REVISED	AZMAN

CLIENT: TENAGA NASIONAL BERHAD
CONTRACT NO: TNB 1799/2023
CONTRACTOR: Viscon
SUP./MFR.: Viscon
TITLE: TELECONTROL OVERVIEW DRAWING
CPB WIRING CIRCUIT
SUBSTATION: PENCAWANG ELEKTRIK 11kV
= PE + RTU /
DRAWING NO.: PDSB UPG1818 RIW
REV.: 00
THIS SHEET: 3-11-2A
NEXT SHEET: 3-12-1



NOTE:

ONLY COM A & COM B
1. ENABLE NOT BOTH

EXISTING RTU COMMUNICATION MODULE WIRING - NCC

ENGINEERING SERVICES DISEMAK 21 SEP 2023 DISTRIBUTION NETWORK DIVISION

GURSHARAN SINGH 10 AUG 23 ES DISTRIBUTION NETWORK DIVISION, TNBD

			DATE	17.07.23	CLIENT: TENAGA NASIONAL BERHAD	CONTRACTOR: Viscon	TITLE: TELECONTROL OVERVIEW DRAWING	= PE	+ RTU	/	
			DRAWN	KHAIRI							
			CHECKED	NAZMI	CONTRACT NO: TNB 1799/2023	SUP./MFR.: Viscon	RTU COMMUNICATION MODULE WIRING - NCC	DRAWING NO.: PDSB UPG1818 RIW	REV.: 00	THIS SHEET: 3-12-1	
REV.	DATE	DRN/CHK/REV	REVISED	AZMAN			SUBSTATION: PENCAWANG ELEKTRIK 11kV			NEXT SHEET: 3-13-1	

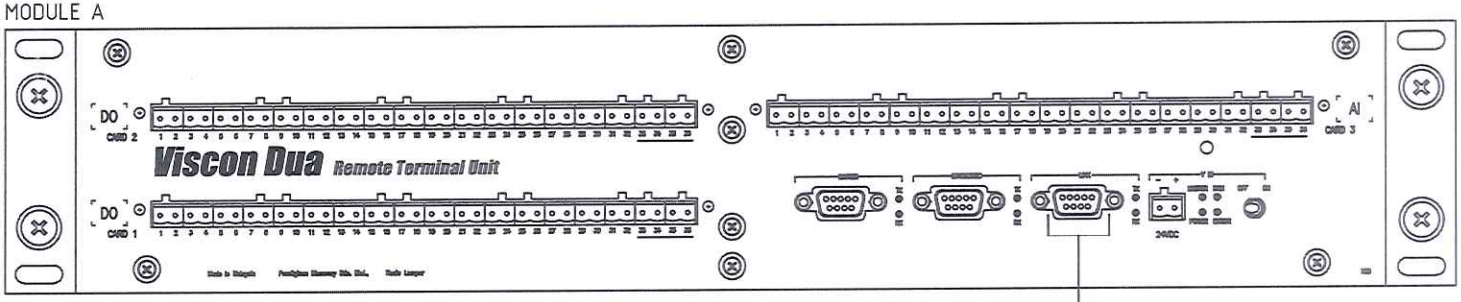
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			DATE	17.07.23	CLIENT: TENAGA NASIONAL BERHAD	CONTRACTOR: Viscon	TITLE: TELECONTROL OVERVIEW DRAWING	= PE + RTU /	
			DRAWN	KHAIRI					
			CHECKED	NAZMI	CONTRACT NO: TNB 1799/2023	SUP./MFR.: Viscon	RTU COMMUNICATION WIRING - IED AND SATELLITE RTU	DRAWING NO.: PDSB UPG1818 RIW	REV.: 00
REV.	DATE	DRN/CHK/REV	REVISED	AZMAN			SUBSTATION: PENCAWANG ELEKTRIK 11kV		THIS SHEET: 3-13-1
									NEXT SHEET: 3-14-1

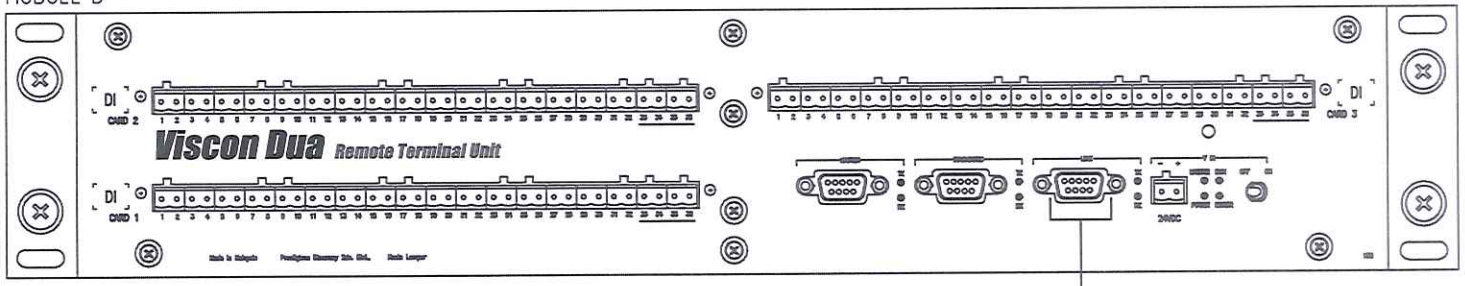
MODULE A



A/LINK B

9-CORE RS422 CABLE

MODULE B

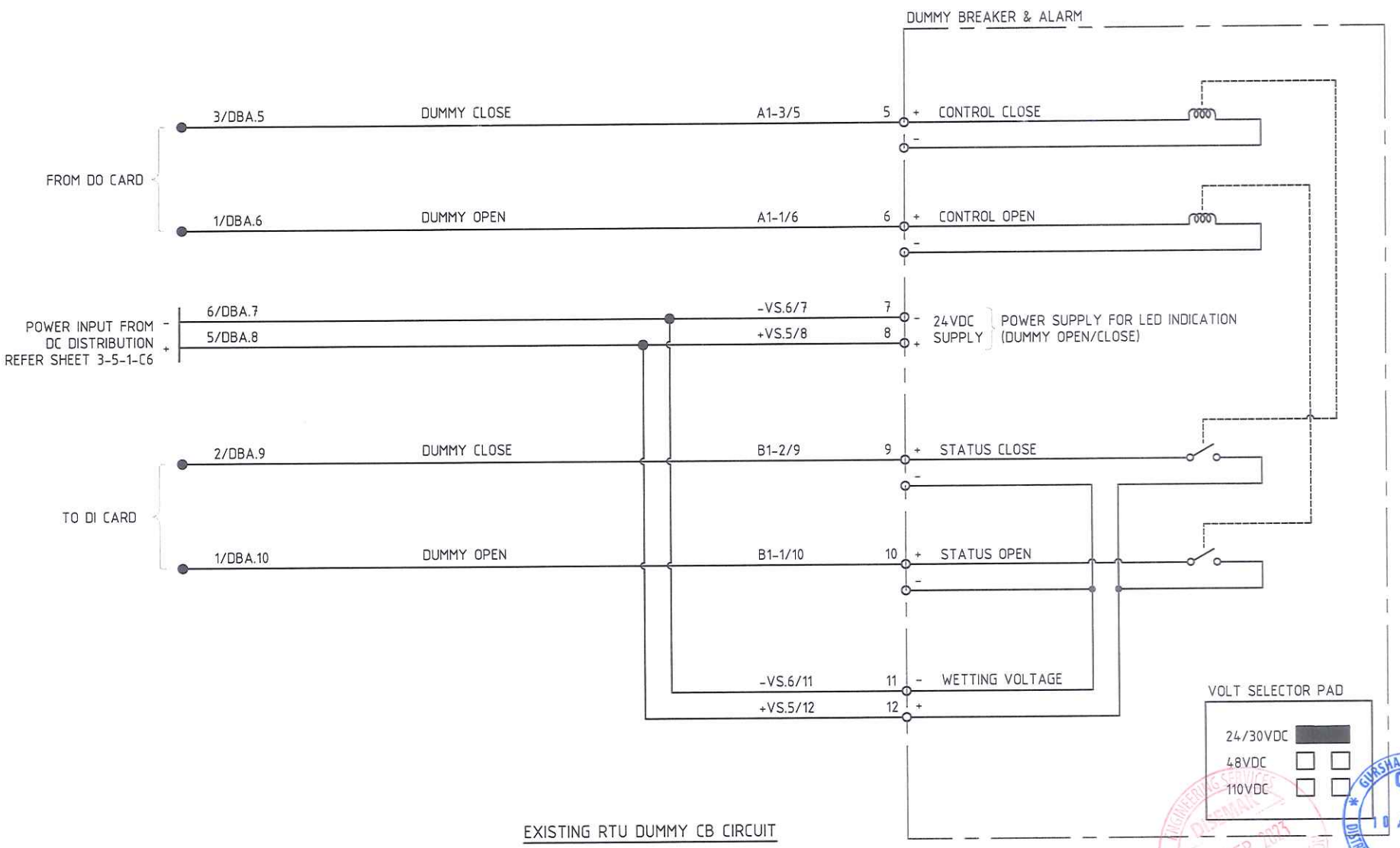


B/LINK A

ENGINEERING SERVICES
DISENAK
21 SEP 2023
GURSHARAN SINGH
10 AUG 23
ES
DISTRIBUTION NETWORK DIVISION

EXISTING RTU INTERNALSUBRACKS / INTER MODULE WIRING (VCB TYPE) - CABINET

			DATE	17.07.23	CLIENT: TENAGA NASIONAL BERHAD	CONTRACTOR: Viscon	TITLE: TELECONTROL OVERVIEW DRAWING	= PE	+ RTU /
			DRAWN	KHAIRI			RTU INTERNAL SUBRACKS / INTER MODULE WIRING - CABINET		
			CHECKED	NAZMI	CONTRACT NO: TNB 1799/2023	SUP./MFR.: Viscon		DRAWING NO.: PDSB UPG1818 RIW	REV.: 00
REV.	DATE	DRN/CHK/REV	REVISED	AZMAN			SUBSTATION: PENCAWANG ELEKTRIK 11kV		THIS SHEET: 3-14-1 / NEXT SHEET: 3-15-1



EXISTING RTU DUMMY CB CIRCUIT

REV.	DATE	DRN/CHK/REV		
			DATE	17.07.23
			DRAWN	KHAIRI
			CHECKED	NAZMI
			REVISED	AZMAN

CLIENT: TENAGA NASIONAL BERHAD

CONTRACT NO: TNB 1799/2023

CONTRACTOR: Viscon

SUP./MFR.: Viscon

TITLE: TELECONTROL OVERVIEW DRAWING

RTU DUMMY CB CIRCUIT

SUBSTATION: PENCAWANG ELEKTRIK 11kV

= PE | + RTU | /

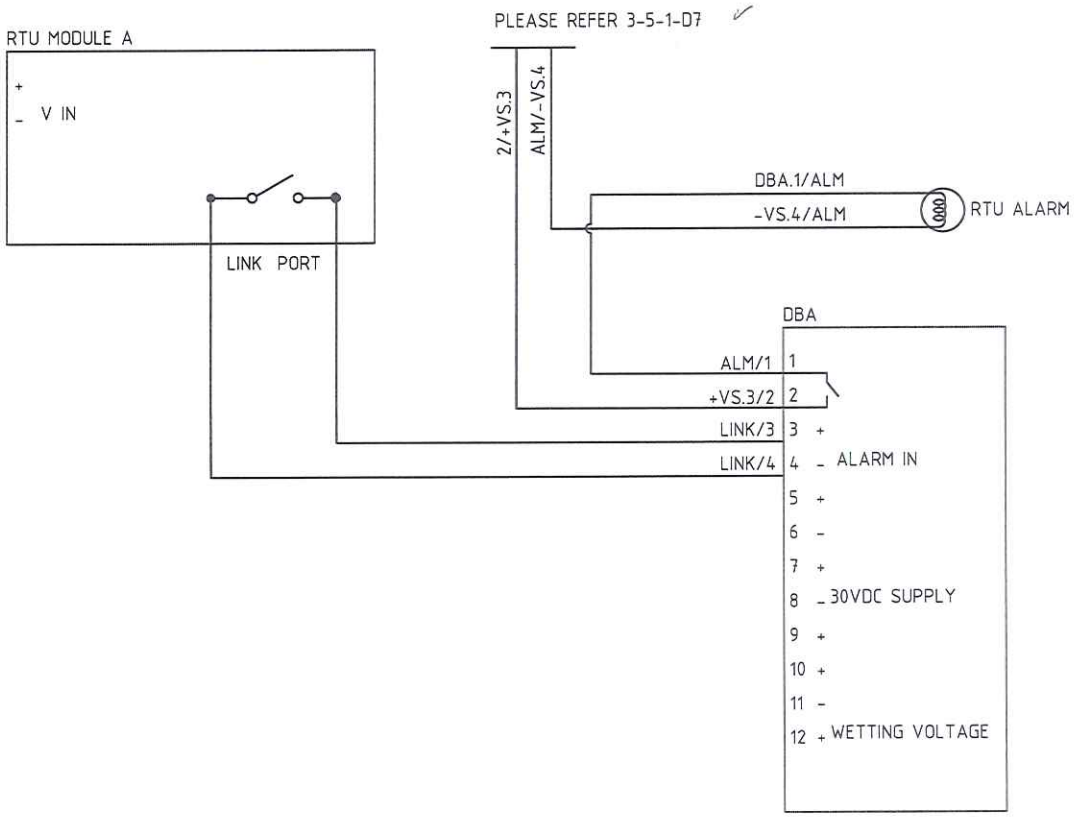
DRAWING NO.: PDSB UPG1818 RIW

REV.: 00

THIS SHEET: 3-15-1

NEXT SHEET: 3-16-1

ENGINEERING SERVICES DIVISION 21 SEP 2023
GURSHARAN SINGH DISTRIBUTION NETWORK DIVISION TNB 10 AUG 23 ES

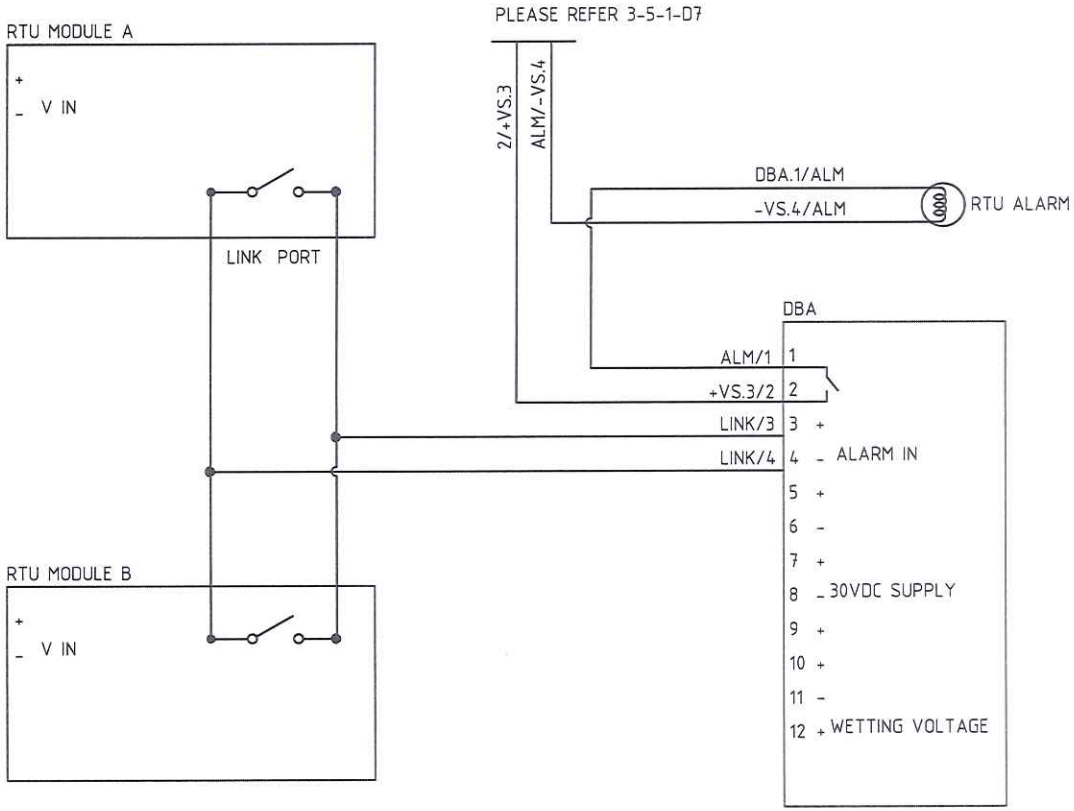


EXISTING RTU FAULT ALARM CIRCUIT (RMU TYPE)

ENGINEERING SERVICES DISEMAK 21 SEP 2023 DISTRIBUTION NETWORK DIVISION

GURSHARAN SINGH 10 AUG 23 ES DISTRIBUTION NETWORK DIVISION

			DATE	17.07.23	CLIENT: TENAGA NASIONAL BERHAD	CONTRACTOR: Viscon	TITLE: TELECONTROL OVERVIEW DRAWING	= PE	+ RTU /
			DRAWN	KHAIRI					
			CHECKED	NAZMI	CONTRACT NO: TNB 1799/2023	SUP./MFR.: Viscon	RTU FAULT ALARM CIRCUIT	DRAWING NO.: PDSB UPG1818 RIW	REV.: 00
REV.	DATE	DRN/CHK/REV	REVISED	AZMAN			SUBSTATION: PENCAWANG ELEKTRIK 11kV		THIS SHEET: 3-16-1 NEXT SHEET: 3-16-2



EXISTING RTU FAULT ALARM CIRCUIT (VCB TYPE)

ENGINEERING SERVICES DISTRIBUTION NETWORK 21 SEP 2023

GURSHARAN SINGH 10 AUG 23 ES DISTRIBUTION NETWORK DIVISION, TNB

REV.	DATE	DRN/CHK/REV		
			DATE	17.07.23
			DRAWN	KHAIRI
			CHECKED	NAZMI
			REVISED	AZMAN

CLIENT: TENAGA NASIONAL BERHAD

CONTRACT NO: TNB 1799/2023

CONTRACTOR: Viscon

SUP./MFR.: Viscon

TITLE: TELECONTROL OVERVIEW DRAWING

RTU FAULT ALARM CIRCUIT

SUBSTATION: PENCAWANG ELEKTRIK 11kV

= PE | + RTU | /

DRAWING NO.: PDSB UPG1818 RIW

REV.: 00

THIS SHEET: 3-16-2

NEXT SHEET: 00